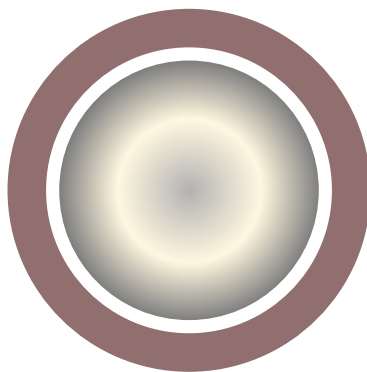


Question:	1	2	Total
Points:	50	50	100
Score:			

1. (50 points) 本題參考 (Wie Bong, *Space Vehicle Dynamics and Control*, Prob. 6.8), 原題想藉由簡單的模型證明有能量損耗時, 除了繞最大慣量主軸外, 繞其餘 2 個主軸的旋轉都不穩定。我懷疑這個目標能否實現, 適當修改原題, 讓大家集思廣義。



整個系統包含二個剛體：外層剛體和內部球體。兩者的質心重合，可相對轉動。外層剛體的主軸慣量分別為 (I_1, I_2, I_3) ，球體的轉動慣量為 I_s 。因為球體為等向性，採用外層剛體的主軸做為座標系。外層剛體相對於主軸座標系的角速度為 $(\omega_1, \omega_2, \omega_3)$ ，球體的角速度為 $(\omega_1 + \sigma_1, \omega_2 + \sigma_2, \omega_3 + \sigma_3)$ 。外層剛體和球體間有一層油膜。因為兩者的轉速不同，油膜產生的黏滯力會加速轉速慢的一方，而降速快的一方。舉例而言，當 $\sigma_1 > 0$ 時，外層剛體在主軸 $\hat{\mathbf{p}}_1$ 的方向受到 $\mu\sigma_1$ 的力矩，而球受到 $-\mu\sigma_1$ 的力矩。

- (a) 請證明系統的運動方程式可表為（注意外層剛體的主軸座標系，也是內部球體的主軸座標系，但不是附體座標系。）

$$\begin{aligned}
 I_1 \dot{\omega}_1 &= (I_2 - I_3) \omega_2 \omega_3 + \mu \sigma_1 \\
 I_2 \dot{\omega}_2 &= (I_3 - I_1) \omega_3 \omega_1 + \mu \sigma_2 \\
 I_3 \dot{\omega}_3 &= (I_1 - I_2) \omega_1 \omega_2 + \mu \sigma_3 \\
 I_s (\dot{\omega}_1 + \dot{\sigma}_1) &= -\mu \sigma_1 + I_s \omega_3 \sigma_2 - I_s \omega_2 \sigma_3 \\
 I_s (\dot{\omega}_2 + \dot{\sigma}_2) &= -I_s \omega_3 \sigma_1 - \mu \sigma_2 + I_s \omega_1 \sigma_3 \\
 I_s (\dot{\omega}_3 + \dot{\sigma}_3) &= I_s \omega_2 \sigma_1 - I_s \omega_1 \sigma_2 - \mu \sigma_3
 \end{aligned} \tag{1}$$

- (b) 直接數值積分式(1)來檢驗系統的特性。首先測試繞最大轉動慣量軸旋轉的穩定性。令 $\mathbf{H}$ 代表系統（外層剛體 + 球）的總角動量、 T 代表總動能。選定適當的參數值，令初始轉速主要在最大轉動慣量軸的方向，其它二主軸向的轉速有微小擾動。依 $\mu = 0$ 和 $\mu \neq 0$ 兩種情形，積分適當時間後，畫出 $\|\mathbf{H}\|$ 、 T 、 ω_1 、 ω_2 、 ω_3 ，隨時間變化的情形。

- (c) 同上小題，但考慮繞中間轉動慣量軸旋轉的穩定性。
- (d) 同上小題，但考慮繞最小轉動慣量軸旋轉的穩定性。
- (e) 由式(1)，證明平衡解存在的必要條件為 $\sigma_1^2 + \sigma_2^2 + \sigma_3^2 = 0$ 。換言之， $\sigma_1 = \sigma_2 = \sigma_3 = 0$ 。進一步可以推論平衡解只能是繞著某個主軸的自旋。
- (f) 接著用解析的方式判別繞主軸旋轉的穩定。以主軸 1 為例，假設平衡解為

$$(\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3)_{\text{equil}} = (\Omega, 0, 0, 0, 0, 0) \quad (2)$$

為了減少變數，先將(1)無因次化。令

$$t^* = \Omega t, \omega_i = \omega_i^* \Omega, \sigma_i = \sigma_i^* \Omega, I_i = I_i^* I_s, \mu^* = \mu / (I_s \Omega) \quad (3)$$

請證明(1)可以表為無因次的形式如下

$$\begin{aligned} I_1 \dot{\omega}_1 &= (I_2 - I_3) \omega_2 \omega_3 + \mu \sigma_1 \\ I_2 \dot{\omega}_2 &= (I_3 - I_1) \omega_3 \omega_1 + \mu \sigma_2 \\ I_3 \dot{\omega}_3 &= (I_1 - I_2) \omega_1 \omega_2 + \mu \sigma_3 \\ (\dot{\omega}_1 + \dot{\sigma}_1) &= -\mu \sigma_1 + \omega_3 \sigma_2 - \omega_2 \sigma_3 \\ (\dot{\omega}_2 + \dot{\sigma}_2) &= -\omega_3 \sigma_1 - \mu \sigma_2 + \omega_1 \sigma_3 \\ (\dot{\omega}_3 + \dot{\sigma}_3) &= \omega_2 \sigma_1 - \omega_1 \sigma_2 - \mu \sigma_3 \end{aligned} \quad (4)$$

注意上式中，為了簡潔我們省略了上標*號，而 $(\dot{\square}) = (d/dt^*)$ 。舉例來說，上式中的 ω_1 即為 ω_1^* 。同時(2)對應的無因次平衡解為

$$(\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3)_{\text{equil}} = (1, 0, 0, 0, 0, 0) \quad (5)$$

- (g) 考慮(5)的穩定性，令

$$(\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3) = (1 + \varepsilon \eta_1, \varepsilon \eta_2, \varepsilon \eta_3, \varepsilon \eta_4, \varepsilon \eta_5, \varepsilon \eta_6) \quad (6)$$

其中 ε 代表微小的量。式(6)代入(4)線性化（忽略 $\mathcal{O}(\varepsilon^2)$ 及更高階項），可以得到 6 條一階的齊次微分方程式（homogeneous equation）。請證明這 6 條方程式可分為 2 組，一組只包含 (η_1, η_4) ，另一組包含 $(\eta_2, \eta_3, \eta_5, \eta_6)$ ，這 2 組方程式都可表為矩陣的形式如下：

$$\mathbf{A} \dot{\boldsymbol{\eta}} = \mathbf{B} \boldsymbol{\eta} \quad (7)$$

上式的一般解為 $\boldsymbol{\eta} = \mathbf{c} e^{st}$ ，其中 $\mathbf{c}$ 為常數向量。代入式(7)化簡得到

$$(s\mathbf{A} - \mathbf{B}) \mathbf{c} = 0$$

由係數矩陣的行列式為 0，

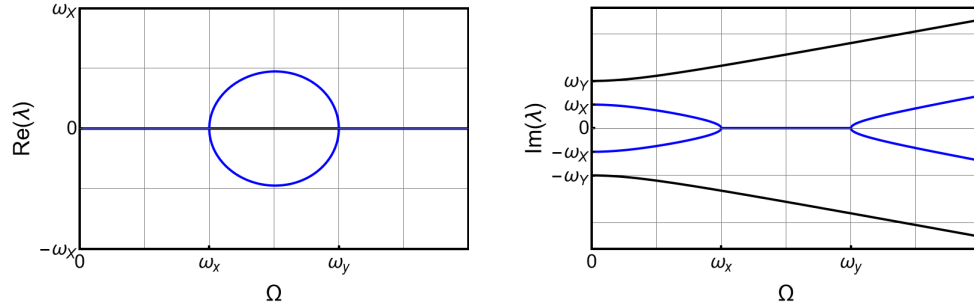
$$\det(s\mathbf{A} - \mathbf{B}) = 0 \quad (8)$$

可解得 (7) 對應的特徵值。如果所有的特徵值的實部都為負，系統為漸近穩定 (asymptotically stable)；如果有一個特徵值的實部為正，系統為不穩定；如果所有特徵值的實部都不為正，系統的穩定性未定，暫且稱為中性穩定 (neutrally stable)。

- (h) 將 (η_1, η_4) 對應的微分方程式表為式 (7) 的形式 (寫出 $\mathbf{A}$, $\mathbf{B}$)，並證明系統為中性穩定。
- (i) 將 $(\eta_2, \eta_3, \eta_5, \eta_6)$ 對應的微分方程式表為式 (7) 的形式 (寫出 $\mathbf{A}$, $\mathbf{B}$)。請試著用解析的方式 (例如自動控制學過的 Routh-Hurwitz criterion) 推論這個系統的穩定性。我個人覺得不容易，不過這不妨礙大家試試。
- (j) 如果解析的方式行不通，可以採用數值方法，令

$$I_2 = 2, I_3 = 1, \mu = 0.1$$

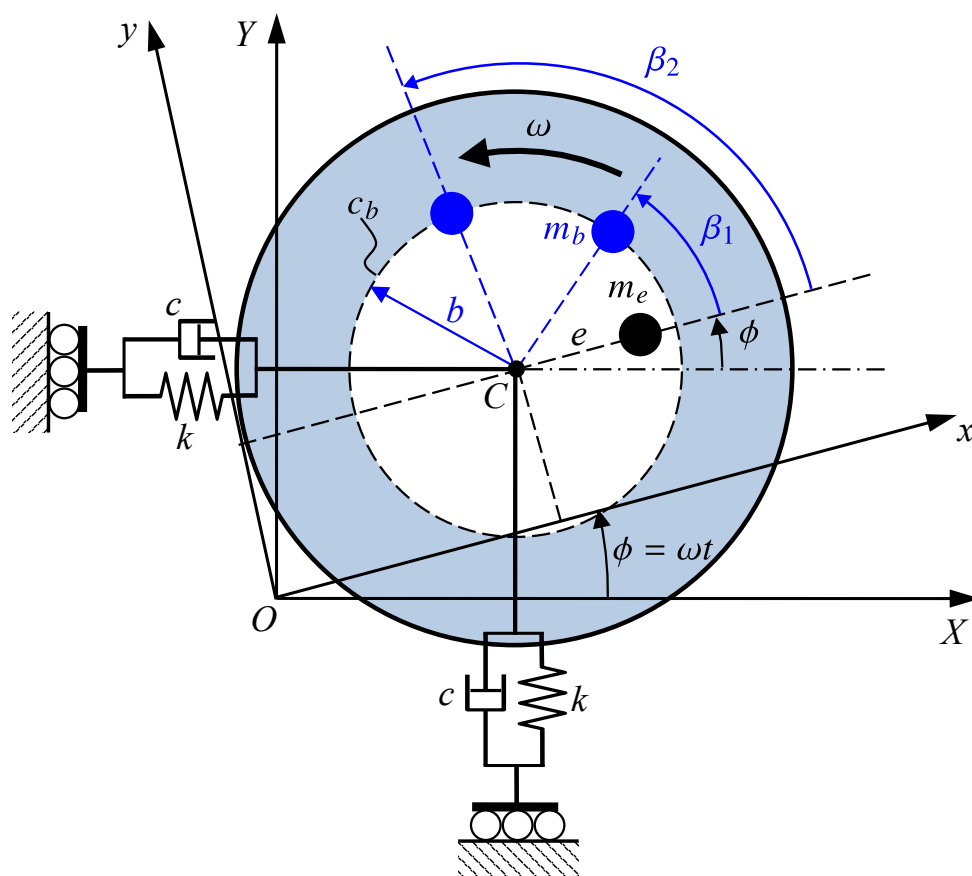
如此給定 I_1 值後，可由式 (8) 解得 4 個特徵值 (可能為實根或共軛複根)。請畫出特徵值的實部和虛部隨著 I_1 變化的情形 (I_1 的範圍為 $I_2 - I_3 < I_1 < I_2 + I_3$)。你的結果應該和下圖類似 (下圖和本題無關)。注意 4 個特徵值不見得會有 4 條線，如果是 2 組共軛複數的話，實部就只有 2 條線；如果有重根，線的數目也會不同。請分別討論 $\mu = 0, 0.1$ 兩種情形，並評論你得到的結果。



2. (50 points) 參考下圖，質量為 M 的均質圓盤以等速率 ω 自轉。圓盤受到等向性支撐，支撐的效應可用水平、垂直方向各一組的彈簧-黏滯阻尼器代表。彈簧的彈性係數為 k ，黏滯阻尼的黏滯係數為 c 。為了模擬偏心旋轉的效應，假設有一個質量為 m_e 的質點固定在圓盤上，隨圓盤轉動。偏心質點和圓盤中心 C 的距離為 e 。此外，圓盤上有一個半徑為 b 的圓形導軌，導軌面有阻尼液，其上有兩顆質量都為 m_b 的滾珠可沿著導軌自由移動。

為了分析方便，採用如下圖所示的座標系，其中，旋轉座標系 (xy 座標系) 和固定座標系 (XY 座標系) 的原點 O 重合。旋轉座標系和圓盤的旋轉速率相同，因此，任何時刻 x 軸和 X 軸的夾角為 $\phi = \omega t$ 。時刻 $t = 0$ 時二者重合。圓盤沒有旋轉時，形心 C 位於原點 O ，偏心質量 m_e 位於 x 軸 (或 X 軸) 上，此時，支撐彈簧沒有變形。圓盤轉動後，因為偏心質量的離心力，圓盤的形心 C 離開原點 O 。任何時刻，

偏心質量和形心 C 的連線都平行 x 軸。以偏心質量和形心 C 的連線為基準，二顆滾珠的分別位於 β_1 和 β_2 處。



選用圓盤形心 C 相對於 xy 座標系的座標 (x, y) ，二顆滾珠的角度 β_1 、 β_2 做為廣義座標。為了處理阻尼力方便，引入 Rayleigh's 耗能函數，

$$R = \frac{c}{2} \dot{X}^2 + \frac{c}{2} \dot{Y}^2 + \frac{c_b}{2} (b\dot{\beta}_1)^2 + \frac{c_b}{2} (b\dot{\beta}_2)^2$$

其中 (X, Y) 為形心 C 相對於固定座標系的座標。系統的彈性能為

$$V = \frac{1}{2} k (x^2 + y^2)$$

系統的運動方程式可由下列形式的 Lagrange 方程式求得

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} + \frac{\partial R}{\partial \dot{q}_i} = 0$$

- 請寫出系統的動能 T 。
- 請寫出系統的運動方程式（需先將 R 表為 $x, y, \dot{x}, \dot{y}, \dot{\phi}$ 的函數）
- 請求出所有可能的平衡解（不隨時間變化的解），及解存在的條件（將 (x, y) 換為極座標 (r, θ) 分析可能較為便利）。
- 令 $\omega_n = \sqrt{k/m}$ 代表系統的自然頻率， $r = \sqrt{x^2 + y^2}$ 代表圓盤的振幅。自選系統參數和初始條件，討論低臨界轉速（ $\omega < \omega_n$ ）和過臨界轉速（ $\omega > \omega_n$ ）下，

r, β_1, β_2 隨時間變化的情形，運動會趨近平衡解嗎？平衡解和 $2m_b b$ 和 $m_e e$ 的比值有關嗎？請適當的評論。

(e) 除了平衡解外，系統長時間後是否存在其它形式的運動，例如週期性運動。

(a) 請證明系統的運動方程式可表為（注意外層剛體的主軸座標系，也是內部球體的主軸座標系，但不是附體座標系。）

$$\begin{aligned}
 I_1 \dot{\omega}_1 &= (I_2 - I_3) \omega_2 \omega_3 + \mu \sigma_1 \\
 I_2 \dot{\omega}_2 &= (I_3 - I_1) \omega_3 \omega_1 + \mu \sigma_2 \\
 I_3 \dot{\omega}_3 &= (I_1 - I_2) \omega_1 \omega_2 + \mu \sigma_3 \\
 I_s (\dot{\omega}_1 + \dot{\sigma}_1) &= -\mu \sigma_1 + I_s \omega_3 \sigma_2 - I_s \omega_2 \sigma_3 \\
 I_s (\dot{\omega}_2 + \dot{\sigma}_2) &= -I_s \omega_3 \sigma_1 - \mu \sigma_2 + I_s \omega_1 \sigma_3 \\
 I_s (\dot{\omega}_3 + \dot{\sigma}_3) &= I_s \omega_2 \sigma_1 - I_s \omega_1 \sigma_2 - \mu \sigma_3
 \end{aligned} \tag{1}$$

令外層剛體主軸座標為 $\{\hat{p}_1, \hat{p}_2, \hat{p}_3\}$

代號 $\begin{cases} \text{外層剛體} = 0 \\ \text{內層剛體} = S \end{cases}$

$$\vec{\omega}_0 = \omega_1 \hat{p}_1 + \omega_2 \hat{p}_2 + \omega_3 \hat{p}_3, [I_0]_{\{\hat{p}\}} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

$$\vec{H}_0 = [I_0]_{\{\hat{p}\}} \vec{\omega}_0 = I_1 \omega_1 \hat{p}_1 + I_2 \omega_2 \hat{p}_2 + I_3 \omega_3 \hat{p}_3$$

$$\vec{M}_0 = \dot{\vec{H}}_0 = \left(\frac{\delta}{\delta t} \vec{H}_0 \right)_{\text{rel}\{\hat{p}\}} + \vec{\omega}_0 \times \vec{H}_0$$

$$= I_1 \dot{\omega}_1 \hat{p}_1 + I_2 \dot{\omega}_2 \hat{p}_2 + I_3 \dot{\omega}_3 \hat{p}_3 + \begin{vmatrix} \hat{p}_1 & \hat{p}_2 & \hat{p}_3 \\ \omega_1 & \omega_2 & \omega_3 \\ I_1 \omega_1 & I_2 \omega_2 & I_3 \omega_3 \end{vmatrix}$$

$$\begin{aligned}
 &= \left(I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 \right) \hat{p}_1 \\
 &\quad + \left(I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_1 \omega_3 \right) \hat{p}_2 \\
 &\quad + \left(I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 \right) \hat{p}_3
 \end{aligned}$$

$$\begin{aligned}
 \text{且 } \vec{M}_0 &= \mu \sigma_1 \hat{p}_1 + \mu \sigma_2 \hat{p}_2 + \mu \sigma_3 \hat{p}_3 \\
 &= \mu \vec{\sigma} \quad (\vec{\sigma} = \sigma_1 \hat{p}_1 + \sigma_2 \hat{p}_2 + \sigma_3 \hat{p}_3)
 \end{aligned}$$

$$\text{又} \begin{cases} I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 = \mu \sigma_1 \\ I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_1 \omega_3 = \mu \sigma_2 \\ I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 = \mu \sigma_3 \end{cases}$$

$$\Rightarrow \begin{cases} I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 + \mu \sigma_1 \\ I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_1 \omega_3 + \mu \sigma_2 \\ I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 + \mu \sigma_3 \end{cases} \quad \text{X}$$

$$\vec{\omega}_S = \vec{\omega}_0 + \vec{\sigma}$$

$$[I_S]_{\{\hat{p}\}} = \begin{bmatrix} I_S & 0 & 0 \\ 0 & I_S & 0 \\ 0 & 0 & I_S \end{bmatrix} = I_S [I_{3 \times 3}]$$

$$\begin{aligned} \vec{H}_S &= [I_S]_{\{\hat{p}\}} \vec{\omega}_S = I_S [I_{3 \times 3}] (\vec{\omega}_0 + \vec{\sigma}) \\ &= I_S (\vec{\omega}_0 + \vec{\sigma}) \end{aligned}$$

$$\begin{aligned} \Rightarrow \vec{M}_S &= \dot{\vec{H}}_S = I_S \left(\left(\frac{\delta}{\delta t} \vec{\omega}_0 \right)_{\text{rel} \{\hat{p}\}} + \left(\frac{\delta}{\delta t} \vec{\sigma} \right)_{\text{rel} \{\hat{p}\}} \right) + I_S \vec{\omega}_0 \times (\vec{\omega}_0 + \vec{\sigma}) \\ &= I_S (\dot{\omega}_1 + \dot{\sigma}_1) \hat{p}_1 + (\dot{\omega}_2 + \dot{\sigma}_2) \hat{p}_2 + (\dot{\omega}_3 + \dot{\sigma}_3) \hat{p}_3 \\ &\quad + I_S \vec{\omega}_0 \times \vec{\sigma} \\ &= I_S ((\dot{\omega}_1 + \dot{\sigma}_1) + \omega_2 \sigma_3 - \omega_3 \sigma_2) \hat{p}_1 \end{aligned}$$

$$+ I_s (\dot{\omega}_2 + \dot{\sigma}_2) + \omega_3 \sigma_1 - \omega_1 \sigma_3) \hat{p}_2 \\ + I_s (\dot{\omega}_3 + \dot{\sigma}_3) + \omega_1 \sigma_2 - \omega_2 \sigma_1) \hat{p}_3$$

$$\text{且 } \vec{M}_s = -\vec{M}_0 = -\mu \sigma_1 \hat{p}_1 - \mu \sigma_2 \hat{p}_2 - \mu \sigma_3 \hat{p}_3$$

$$\text{故} \Rightarrow \begin{cases} I_s (\dot{\omega}_1 + \dot{\sigma}_1) + \omega_2 \sigma_3 - \omega_3 \sigma_2 = -\mu \sigma_1 \\ I_s (\dot{\omega}_2 + \dot{\sigma}_2) + \omega_3 \sigma_1 - \omega_1 \sigma_3 = -\mu \sigma_2 \\ I_s (\dot{\omega}_3 + \dot{\sigma}_3) + \omega_1 \sigma_2 - \omega_2 \sigma_1 = -\mu \sigma_3 \end{cases}$$

$$\text{故} \Rightarrow \begin{cases} I_s (\dot{\omega}_1 + \dot{\sigma}_1) = -I_s \omega_2 \sigma_3 + I_s \omega_3 \sigma_2 - \mu \sigma_1 \\ I_s (\dot{\omega}_2 + \dot{\sigma}_2) = -I_s \omega_3 \sigma_1 + I_s \omega_1 \sigma_3 - \mu \sigma_2 \\ I_s (\dot{\omega}_3 + \dot{\sigma}_3) = -I_s \omega_1 \sigma_2 + I_s \omega_2 \sigma_1 - \mu \sigma_3 \end{cases} \quad \text{X}$$

(b) 直接數值積分式(1)來檢驗系統的特性。首先測試繞最大轉動慣量軸旋轉的穩定性。令 $\mathbf{H}$ 代表系統（外層剛體 + 球）的總角動量、 T 代表總動能。選定適當的參數值，令初始轉速主要在最大轉動慣量軸的方向，其它二主軸向的轉速有微小擾動。依 $\mu = 0$ 和 $\mu \neq 0$ 兩種情形，積分適當時間後，畫出 $\|\mathbf{H}\|$ 、 T 、 ω_1 、 ω_2 、 ω_3 ，隨時間變化的情形。

(c) 同上小題，但考慮繞中間轉動慣量軸旋轉的穩定性。

(d) 同上小題，但考慮繞最小轉動慣量軸旋轉的穩定性。

$$\vec{H}_{total} = \sum_{\bar{i}=1}^3 (I_{\bar{i}} \omega_{\bar{i}} + I_s (\omega_{\bar{i}} + \sigma_{\bar{i}})) \hat{p}_{\bar{i}}$$

$$\Rightarrow \|\vec{H}_{total}\| = \sqrt{\sum_{\bar{i}=1}^3 (I_{\bar{i}} \omega_{\bar{i}} + I_s (\omega_{\bar{i}} + \sigma_{\bar{i}}))^2}$$

$$T_{total} = \frac{1}{2} \vec{\omega}_0 \cdot [I_0]_{\{\hat{p}\}} \vec{\omega}_0 + \frac{1}{2} m (\dot{V}_0)_0^2 + \frac{1}{2} \vec{\omega}_s \cdot [I_s]_{\{\hat{p}\}} \vec{\omega}_s + \frac{1}{2} m (\dot{V}_s)_s^2 \\ = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) + \frac{1}{2} (\vec{\omega}_0 + \vec{\sigma}) \cdot I_s (\vec{\omega}_0 + \vec{\sigma})$$

$$T_{total} = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2 + \frac{1}{2} I_s \sum_{i=1}^3 (\omega_i + \sigma_i)^2$$

$$M_{total} = \dot{H}_{total} = 0 \Rightarrow \vec{H}_{total} = \text{const.} \quad (\text{角動量守恆})$$

$$\text{力矩作功率} \sum \frac{d}{dt} (\vec{M} \cdot \vec{\sigma}) = \sum \vec{M} \cdot \vec{\omega} = \text{動能變化率} \quad (\text{功能定理})$$

$$\begin{aligned} \Rightarrow \dot{T} &= \vec{M}_0 \cdot \vec{\omega}_0 + \vec{M}_s \cdot \vec{\omega}_s = \mu \vec{\sigma} \cdot \vec{\omega}_0 + (-\mu \vec{\sigma}) \cdot (\vec{\omega}_0 + \vec{\sigma}) \\ &= -\mu \|\vec{\sigma}\|^2 = -\mu(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) \end{aligned}$$

$$\Rightarrow \dot{T} = -\mu(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) \leq 0 \Rightarrow \begin{cases} \mu = 0 \Rightarrow T = \text{const. 守恆} \\ \mu > 0 \Rightarrow T \text{ 單調遞減} \end{cases}$$

★ $\mu = 0 \Rightarrow$ 無耗能 $\Rightarrow$ 符合中間軸定理

$\mu > 0 \Rightarrow$ 耗能 $\Rightarrow \vec{H}_{total}$ 仍守恆, 但 T_{total} 單調下降 $\Rightarrow$ 長時間後內外殼相對轉率重

$\Rightarrow \vec{\sigma} \rightarrow \vec{0} \Rightarrow$ 內外剛體合一 $\Rightarrow$ 主慣量為 $(I_1 + I_s, I_2 + I_s, I_3 + I_s)$

$\Rightarrow$ 最後純繞 i 軸轉 (H 守恆) $\Rightarrow T_i = \frac{\|\vec{H}\|^2}{2(I_i + I_s)} \propto \frac{1}{I_i + I_s} \quad (\text{for } \vec{\sigma} \rightarrow \vec{0})$

$\Rightarrow$ 若 I_i 愈大 $\Rightarrow T_i$ 愈小 $\Rightarrow$ 故有耗能 sys T_i 下降 $\Rightarrow$ 系統傾向往最大 i 軸吸引轉向

取 $I_1=1, I_2=2, I_3=3, I_s=1$ (以進行模擬)

$\Rightarrow \begin{cases} \hat{p}_1: \text{最小慣量主軸} \\ \hat{p}_2: \text{中間慣量主軸} \\ \hat{p}_3: \text{最大慣量主軸} \end{cases}$

```
q1DefaultParams = <|
  "I" -> {1., 2., 3.},
  [座標單位]
  "Is" -> 1.
  |>

Q1AxisName[axis_Integer] := Switch[axis,
  [整數] [切換]
  1, "minimum axis p1",
  2, "intermediate axis p2",
  3, "maximum axis p3",
  _, "unknown axis"
];
```

$\Rightarrow$ 初始條件: $\begin{cases} \omega_{min}(0) = 1, \text{ 並在另外兩軸加入 } 0.02 \text{ 及 } -0.015 \text{ 之隨機小擾動} \\ \sigma(0) = 0 \end{cases}$

使用 wolfram mathematica 進行模擬

初值台條件 繞主軸轉 另外兩軸曲加入小擾動

```
Q1InitialState[axis_Integer, eps1_ : 0.02, eps2_ : -0.015] := Module[
  {w = {0., 0., 0.}, others},
  w[[axis]] = 1.;  $\Rightarrow \omega_{axis}(0) = 1$ 
  others = Complement[{1, 2, 3}, {axis}];
  w[[others[[1]]] = eps1;
  w[[others[[2]]] = eps2;
  Join[w, {0., 0., 0.}]
];
```

列出式(1)之黑頁函式ODE

```
Q1RHS[muVal_?NumericQ, pars_ : q1DefaultParams][state_List] := Module[
  {I1v, I2v, I3v, Isv, w1v, w2v, w3v, s1v, s2v, s3v,
   wdot, sphereRHS, sigdot},
  {I1v, I2v, I3v} = pars["I"];
  Isv = pars["Is"];
  {w1v, w2v, w3v, s1v, s2v, s3v} = state;  $\Rightarrow$  input state  $\{\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3\}$ 
  wdot = {
    ((I2v - I3v) w2v w3v + muVal s1v) / I1v,
    ((I3v - I1v) w3v w1v + muVal s2v) / I2v,
    ((I1v - I2v) w1v w2v + muVal s3v) / I3v
  };
  sphereRHS = {
    (-muVal s1v + Isv w3v s2v - Isv w2v s3v) / Isv,
    (-Isv w3v s1v - muVal s2v + Isv w1v s3v) / Isv,
    (Isv w2v s1v - Isv w1v s2v - muVal s3v) / Isv
  };
  sigdot = sphereRHS - wdot;
  Join[wdot, sigdot]
];
```

回傳 output $\{\dot{\omega}_1, \dot{\omega}_2, \dot{\omega}_3, \dot{\sigma}_1, \dot{\sigma}_2, \dot{\sigma}_3\}$

$$\dot{\vec{\omega}} = \begin{cases} \dot{\omega}_1 = \left(\frac{I_2 - I_3}{I_1}\right) \omega_2 \omega_3 + \frac{\mu \sigma_1}{I_1} \\ \dot{\omega}_2 = \left(\frac{I_3 - I_1}{I_2}\right) \omega_1 \omega_3 + \frac{\mu \sigma_2}{I_2} \\ \dot{\omega}_3 = \left(\frac{I_1 - I_2}{I_3}\right) \omega_1 \omega_2 + \frac{\mu \sigma_3}{I_3} \end{cases}$$

$\dot{\sigma}$ $\dot{\omega} + \dot{\sigma}$ $\dot{\omega}$
sigdot = sphereRHS - wdot;

$$\begin{cases} (\dot{\omega}_1 + \dot{\sigma}_1) = (-I_3 \omega_2 \sigma_3 + I_3 \omega_3 \sigma_2 - \mu \sigma_1) / I_s \\ (\dot{\omega}_2 + \dot{\sigma}_2) = (-I_3 \omega_3 \sigma_1 + I_3 \omega_1 \sigma_3 - \mu \sigma_2) / I_s \\ (\dot{\omega}_3 + \dot{\sigma}_3) = (-I_3 \omega_1 \sigma_2 + I_3 \omega_2 \sigma_1 - \mu \sigma_3) / I_s \end{cases}$$

Sphere RHS

做數值積分 $\Rightarrow axis = 1, 2, 3, \mu = 0.01, t_{max} = \text{積分時間}$

```
Q1Solve[axis_Integer, muVal_?NumericQ,
  {vars, rhs, init},
  vars = {w1, w2, w3, sig1, sig2, sig3};
  rhs = Q1RHS[muVal, pars][Through[vars[t]]];
  init = Thread[Through[vars[0]] == Q1InitialState[axis]];
  NDSolveValue[
    Thread[{w1'[t], w2'[t], w3'[t], sig1'[t], sig2'[t], sig3'[t]} == rhs],
    {t, 0, tmax},
    vars,
    {t, 0, tmax},
    MaxSteps -> Infinity,
    MaxStepFraction -> 1 / 5000,
    AccuracyGoal -> 11,
    PrecisionGoal -> 11
  ]
```

計算數值解下之 H - $\|H\|$

```
Q1HVector[state_List, pars_ : q1DefaultParams] := Module[{Iv, Isv, wv, sigv},
  Iv = pars["I"];
  Isv = pars["Is"];
  wv = state[[1 ;; 3]];
  sigv = state[[4 ;; 6]];
  Iv * wv + Isv (wv + sigv)
];

Q1HNorm[state_List, pars_ : q1DefaultParams] := Norm[Q1HVector[state, pars]];
```

計算數值解下之 T

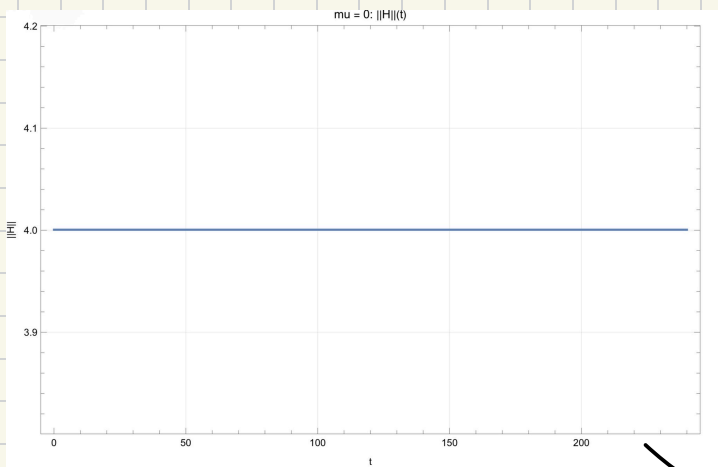
```
Q1Energy[state_List, pars_ : q1DefaultParams] := Module[{Iv, Isv, wv, sigv},
  Iv = pars["I"];
  Isv = pars["Is"];
  wv = state[[1 ;; 3]];
  sigv = state[[4 ;; 6]];
  1 / 2 Total[Iv * wv^2] + 1 / 2 Isv Total[(wv + sigv)^2]
];
```

計算數值解下之 T

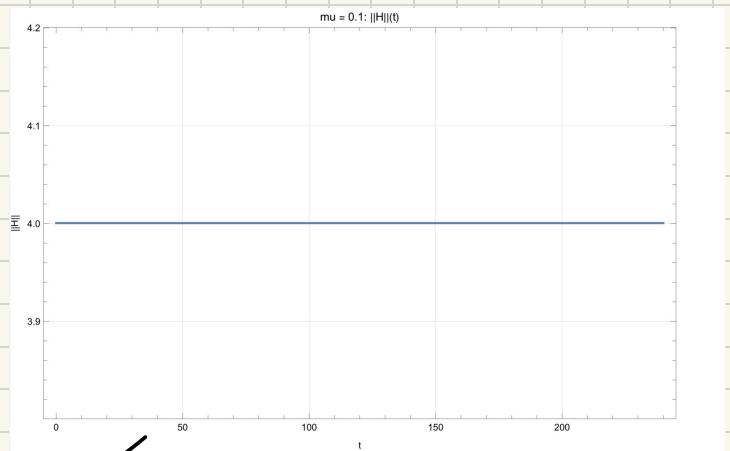
```
Q1EnergyDecayRate[state_List, muVal_?NumericQ] :=
  -muVal Total[state[[4 ;; 6]]^2];
```

畫圖 (b) 繞 P 轉

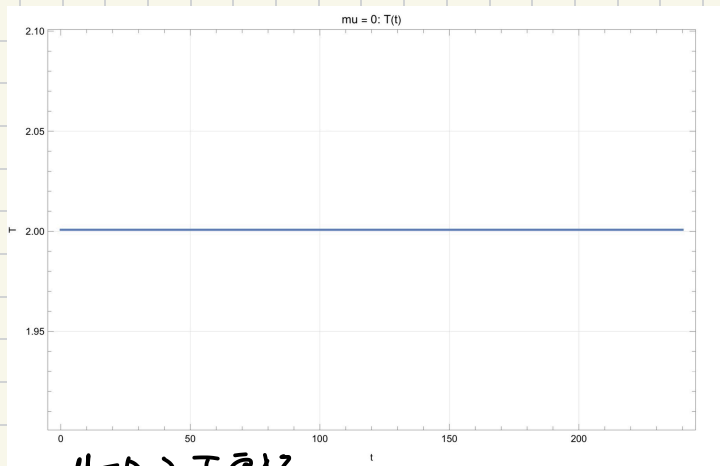
$\mu = 0$ (無耗能系統)



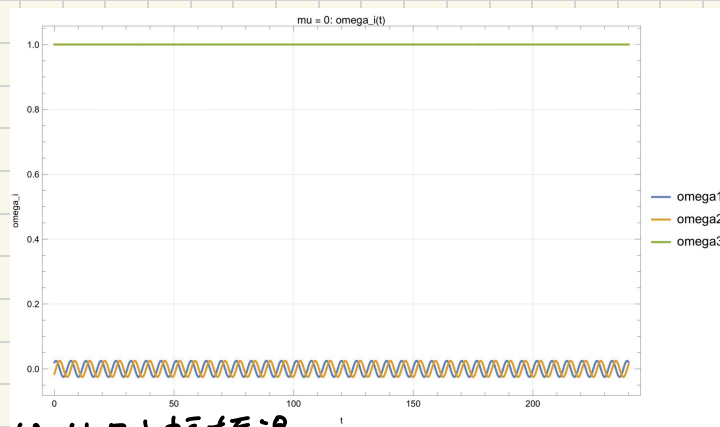
$\mu = 0.1$ (耗能系統)



$\|H\|$ 守恒

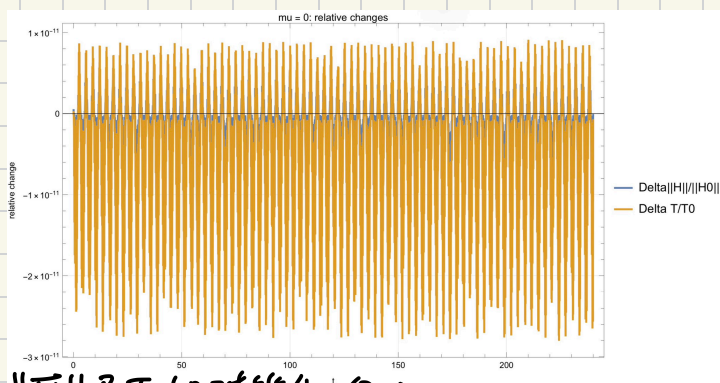


$\mu=0 \Rightarrow T$ 恒定

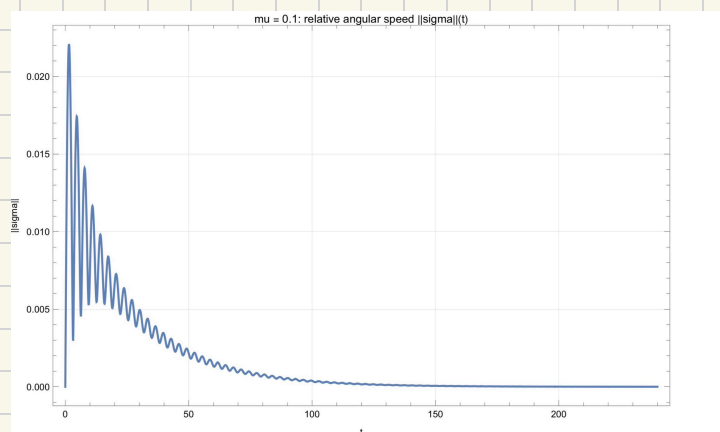


ω_1, ω_2 只小幅振荡, $\omega_3 = \text{const.}$

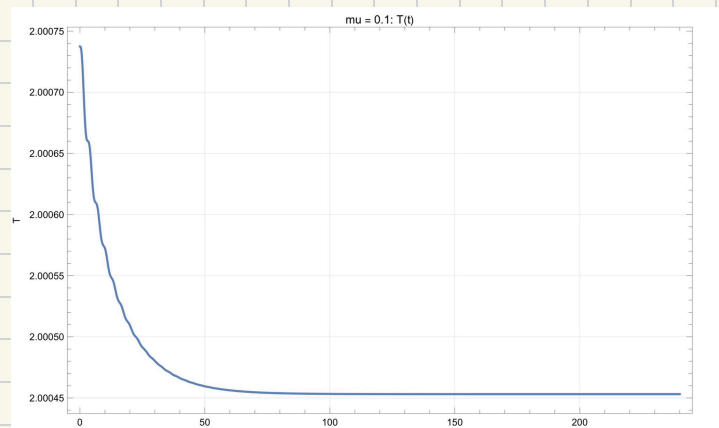
$\Rightarrow$ 无耗能绕最大工轴转稳定必



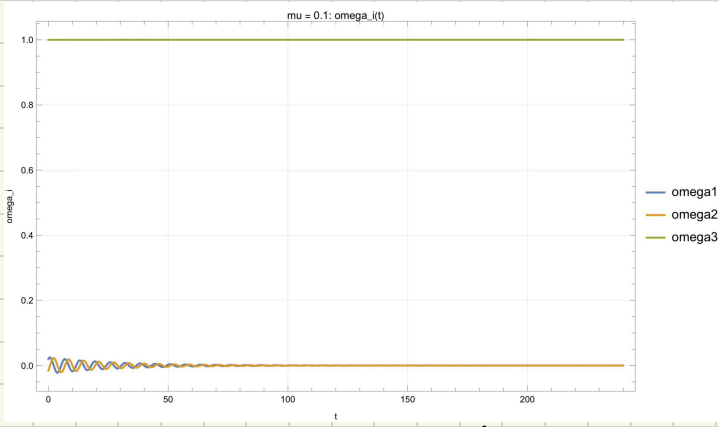
$\|H\|$ 及 T 相对变化很小



$\|\sigma\|$ 最终耗散至 0 (内外合一)

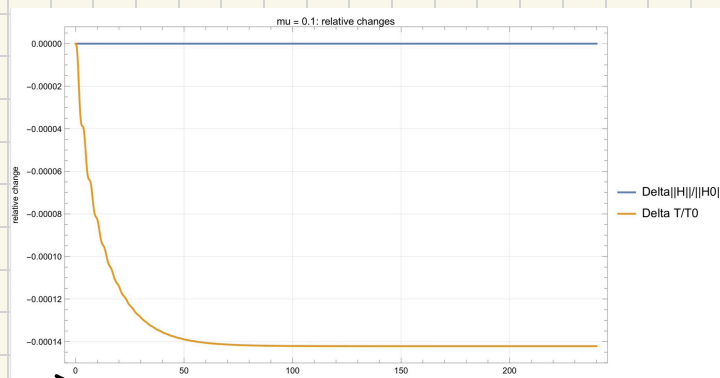


$\mu > 0 \Rightarrow \dot{T} = -\mu \|\sigma\| \leq 0$

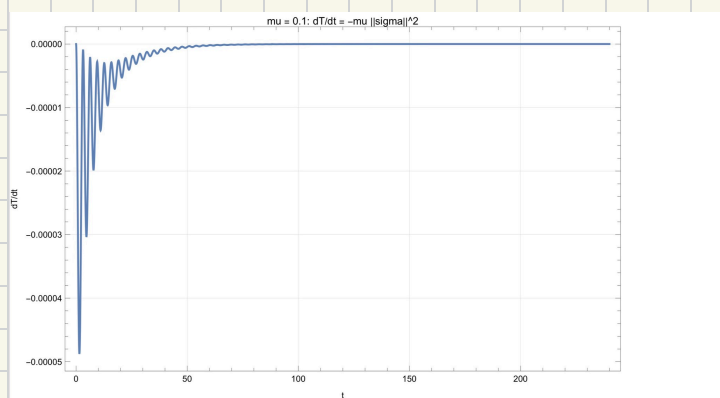


ω_1, ω_2 只小幅振荡后能量耗散 $\rightarrow 0$, $\omega_3 = \text{const.}$

$\Rightarrow$ 有耗能绕最大工轴转稳定必
(耗能 sys 被最大工轴吸引转向)



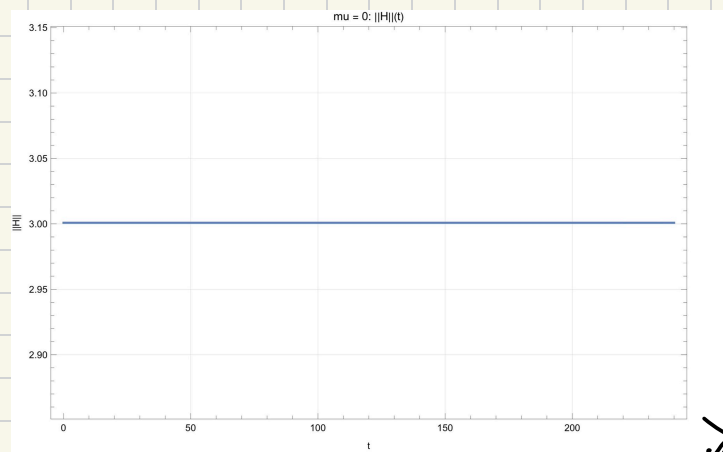
$\|H\|$ 恒定 - T 单调下降



T 始终为负, 但趋缓

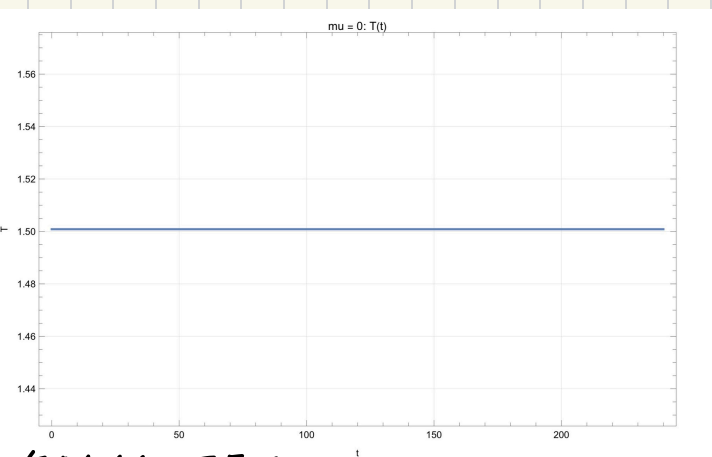
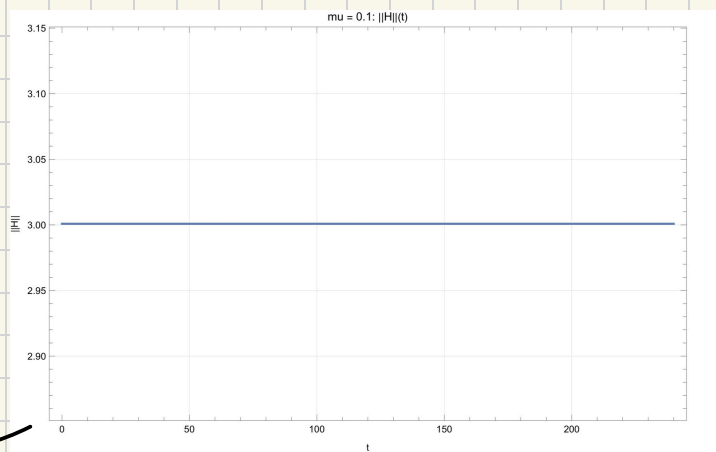
(c) 繞中間軸旋轉

$\mu=0$ (無耗能系統)

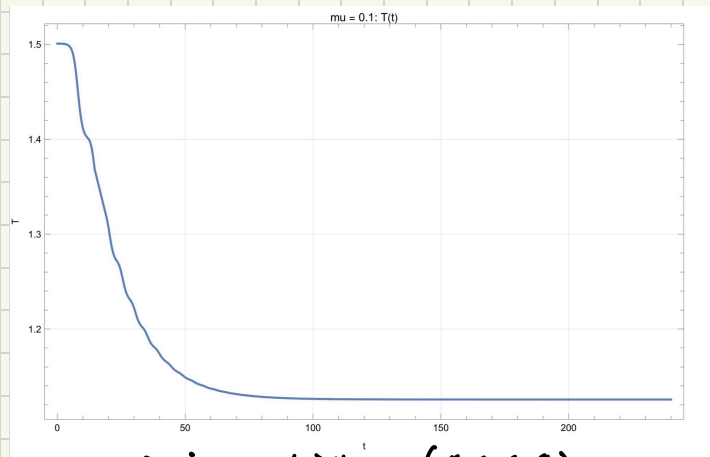


$\|H\|$ 守恒

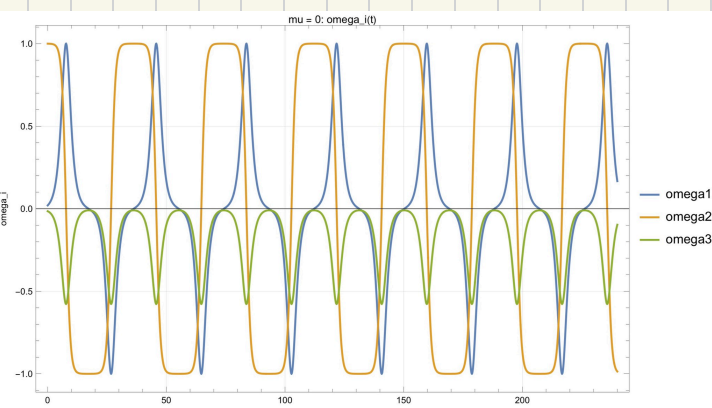
$\mu=0.1$ (耗能系統)



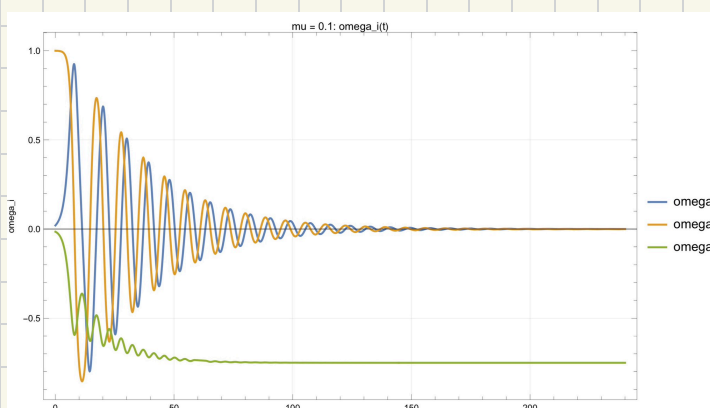
無耗能 $\Rightarrow T$ 守恒



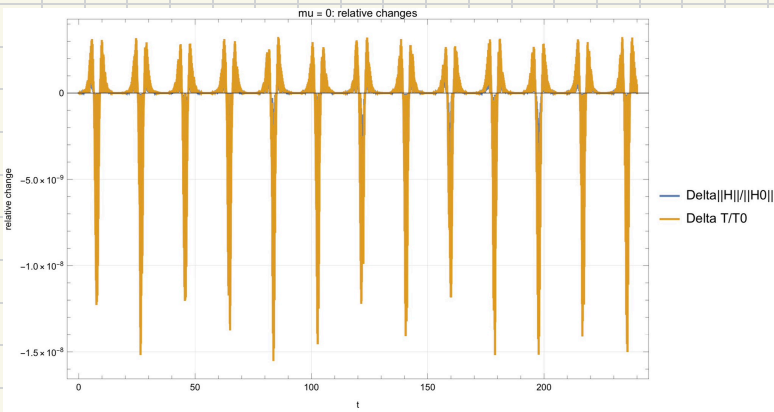
$\mu>0 \Rightarrow \dot{T} = -\mu \| \vec{\omega} \|^2 \leq 0$ (耗能)



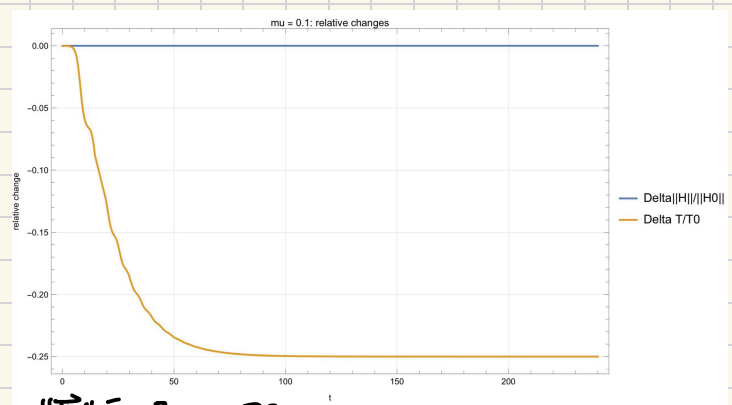
無耗能同中間軸定理, ω_2 反復旋轉不穩定



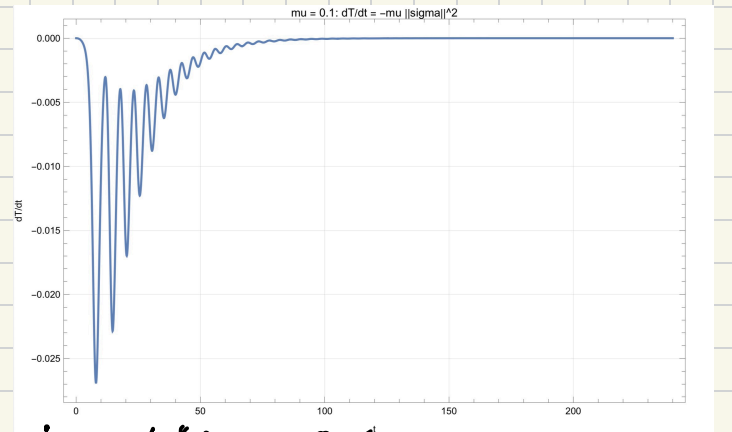
耗能系統 $\Rightarrow \omega_2$ 迅速耗散, 轉動力方向被最大工軸 $\hat{p}_3$ 吸引 $\Rightarrow \omega_3$ 成主旋轉



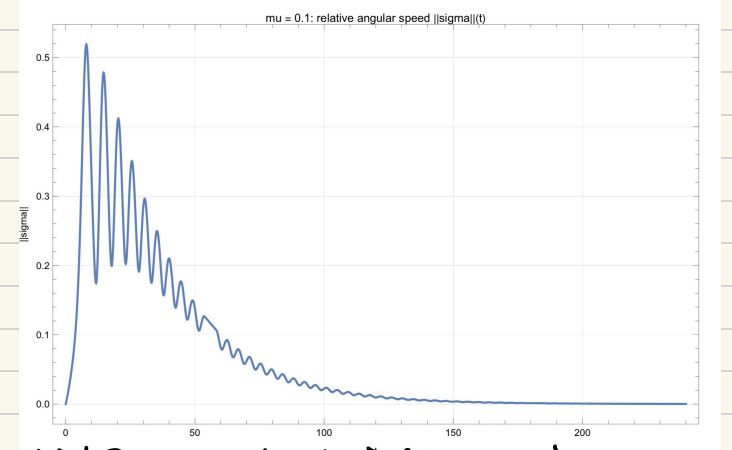
$\|H\|$ 及 T 相对变化很小



$\|H\|$ 守恒, T 单调下降



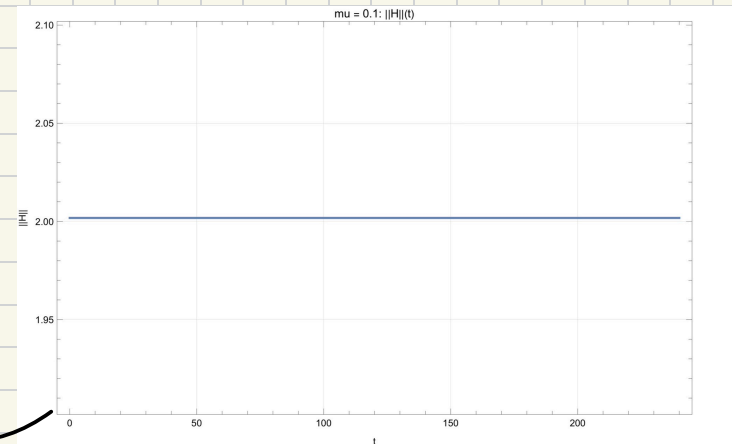
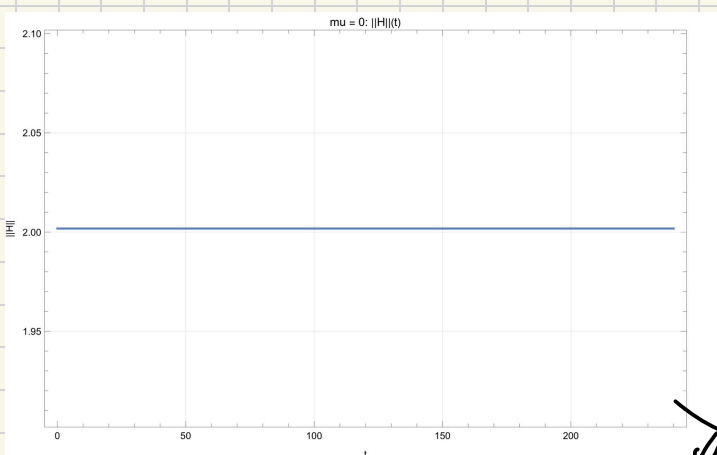
T 始终为负, 但趋缓



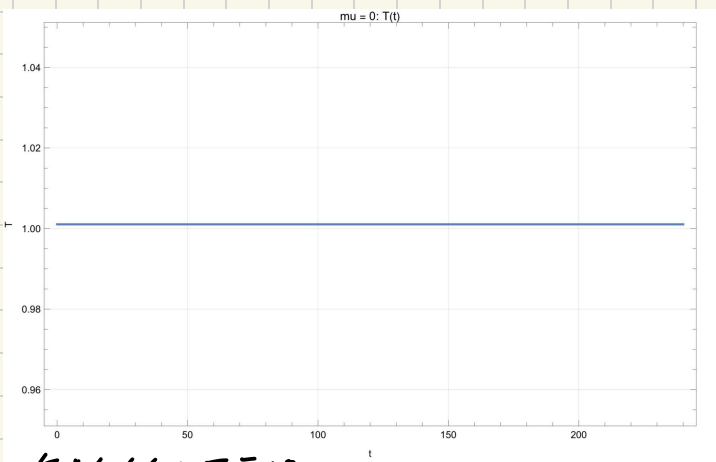
$\|H\|$ 最终耗散至 0 (内外合一)

(d) 绕最小工轴 $\hat{p}$ 旋转

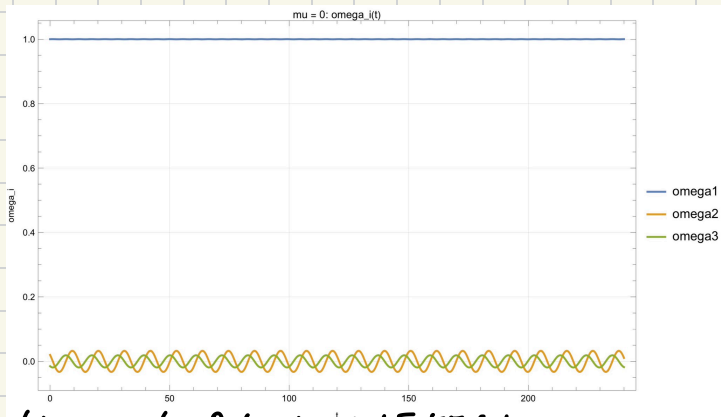
$\mu=0$ (无耗能系统)



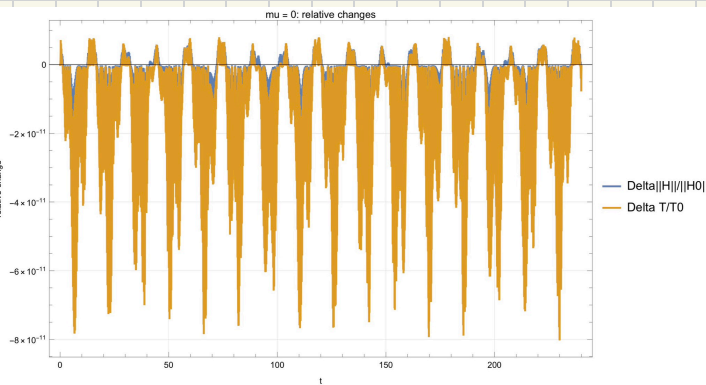
$\|H\|$ 守恒



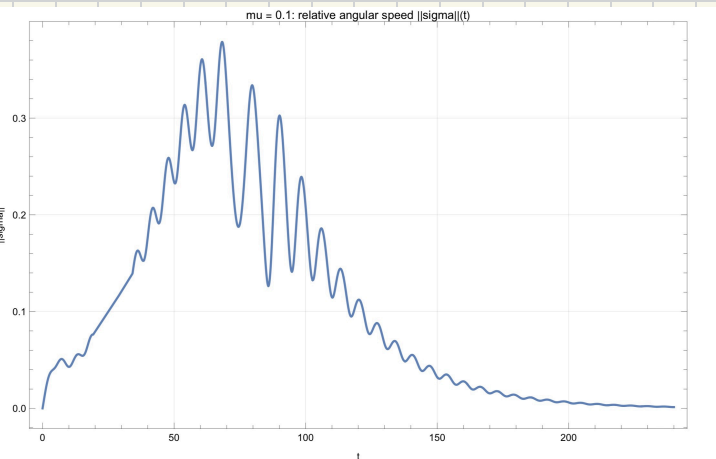
無耗能 $\Rightarrow T$ 可恒



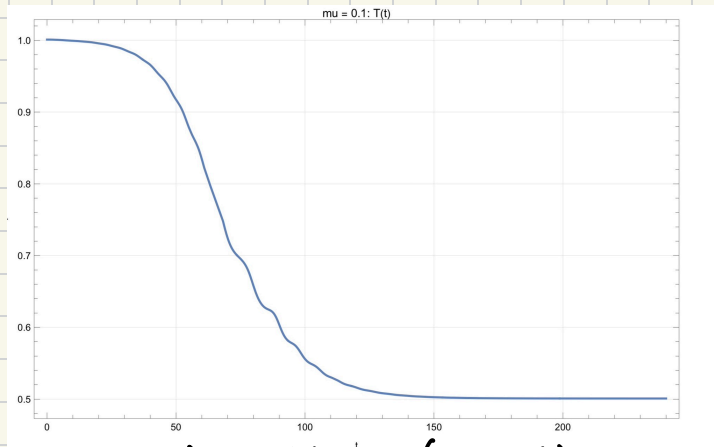
$\omega_1 = \text{const.}$, 且 ω_2, ω_3 小幅度振動
 $\Rightarrow$ 無耗能系統最小工軸穩定



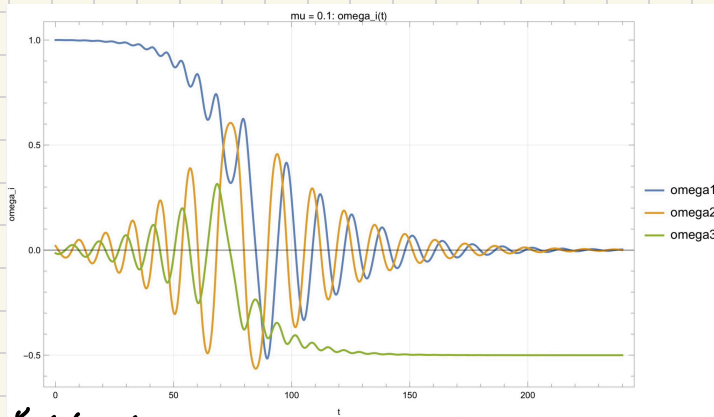
$\|H\|$ 及 T 相對變化很小



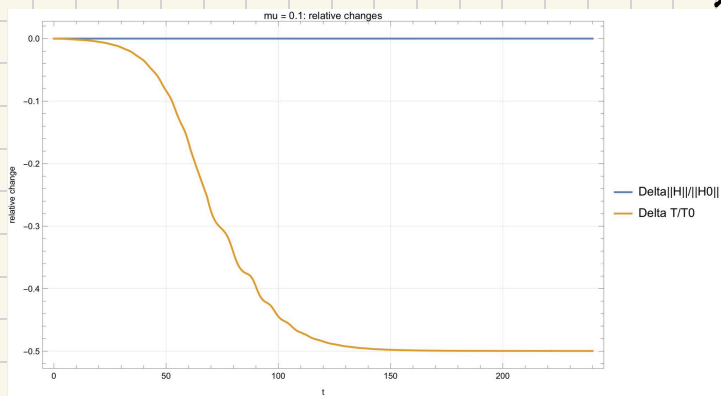
$\|\sigma\|$ 最終耗散至 0 (內外合一)



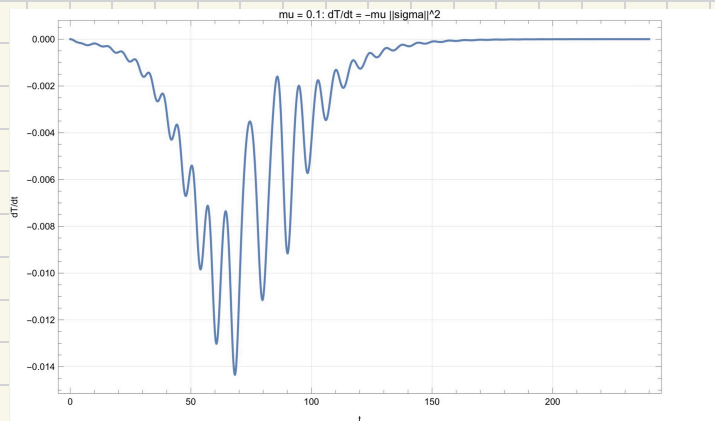
$\mu > 0 \Rightarrow \dot{T} = -\mu \|\sigma\| \leq 0$ (耗能)



耗能系統 $\Rightarrow \omega_1$ 迅速耗散, 轉動力
 方向被最大工軸 $\hat{p}_3$ 吸引 $\Rightarrow \omega_3$ 成主旋轉



$\|H\|$ 可恒 - T 單調下降



$\dot{T}$ 始終為負, 但趨緩

(c) 平衡解 $\Rightarrow$ i.e. $\frac{d}{dt} = 0$

by 式(1) 内球部份 $\Rightarrow I_3 (\overbrace{\vec{\omega}_0 + \vec{\sigma}}^{\neq 0}) + I_3 \vec{\omega}_0 \times \vec{\sigma} = -\mu \vec{\sigma}$

$$\Rightarrow I_3 \vec{\omega}_0 \times \vec{\sigma} = -\mu \vec{\sigma}$$

$$\Rightarrow I_3 \underbrace{(\vec{\omega}_0 \times \vec{\sigma}) \cdot \vec{\sigma}}_{\perp \vec{\sigma}} = -\mu \|\vec{\sigma}\|^2 = 0$$

$$\Rightarrow -\mu(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) = 0$$

① 若 $\mu > 0$ (耗能 sys)

$$\Rightarrow \sigma_1^2 + \sigma_2^2 + \sigma_3^2 = 0 \quad (i.e. \sigma_1 = \sigma_2 = \sigma_3 = 0) \quad (\text{内外各一})$$

$$\Rightarrow \text{式(1) 外層部份} \begin{cases} (I_2 - I_3) \omega_2 \omega_3 = 0 \\ (I_3 - I_1) \omega_1 \omega_3 = 0 \\ (I_1 - I_2) \omega_1 \omega_2 = 0 \end{cases}$$

$$\text{若 } I_1 \neq I_2 \neq I_3 \Rightarrow (\omega_1, \omega_2, \omega_3) = (\Omega, 0, 0) \vee (0, \Omega, 0) \vee (0, 0, \Omega)$$

$\Rightarrow$ i.e. 繞某一主軸自旋

② 若 $\mu = 0$ (無耗能)

$$\Rightarrow I_3 \vec{\omega}_0 \times \vec{\sigma} = 0 \Rightarrow \vec{\omega}_0 \parallel \vec{\sigma} \Rightarrow \ddot{\vec{M}}_0 = 0 = \dot{\vec{H}}_0 = \vec{\omega}_0 \times \vec{H}_0 = \vec{\omega}_0 \times [I_0]_{\text{ref}} \vec{\omega}_0$$

$$\Rightarrow \vec{\omega}_0 \parallel [I_0]_{\text{ref}} \vec{\omega}_0 \Rightarrow [I_0]_{\text{ref}} \vec{\omega}_0 = \lambda \vec{\omega}_0 \quad (\text{eigen vector})$$

$\Rightarrow \vec{\omega}_0$ 為主軸方向 $\Rightarrow \vec{\sigma} \parallel \vec{\omega}_0$ 亦為主軸方向

$$\Rightarrow \text{平衡解可寫} \begin{cases} \vec{\omega}_0 = \Omega \hat{p}_k \\ \vec{\sigma} = S \hat{p}_k \end{cases} \quad (k=1,2,3) \quad (S \text{ 可以 } \neq \Omega)$$

$$(f) \text{ 平衡解} \Rightarrow (\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3)_{\text{equil}} = (\Omega, 0, 0, 0, 0, 0)$$

$$\text{以 } t^* = \Omega t, \omega_i = \Omega \omega_i^*, \sigma_i = \Omega \sigma_i^*, I_i = I_s I_i^*, \mu^* = \frac{\mu}{I_s \Omega} \text{ 無因次化式 (1)}$$

$$\Rightarrow \frac{d\omega_i}{dt} = \Omega^2 \frac{d\omega_i^*}{dt^*}, \quad \frac{d\sigma_i}{dt} = \Omega^2 \frac{d\sigma_i^*}{dt^*}$$

$$I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 + \mu \sigma_1$$

↓ 無因次化

$$I_s \Omega^2 I_1 \dot{\omega}_1 = I_s \Omega^2 (I_2 - I_3) \omega_2 \omega_3 + I_s \Omega^2 \mu \sigma_1$$

↓ $\div I_s \Omega^2$

$$\left\{ \begin{array}{l} I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 + \mu \sigma_1 \\ \quad \downarrow \text{同理} \end{array} \right. \quad \text{✗}$$

$$\left\{ \begin{array}{l} I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_1 \omega_3 + \mu \sigma_2 \\ I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 + \mu \sigma_3 \end{array} \right. \quad \text{✗} \quad \left(\text{省略}^* \text{ 且 } (\dot{}) = \left(\frac{d}{dt^*} \right) \right)$$

$$I_s (\dot{\omega}_1 + \dot{\sigma}_1) = -I_s \omega_2 \sigma_3 + I_s \omega_3 \sigma_2 - \mu \sigma_1$$

↓ 無因次化

$$I_s \Omega^2 (\dot{\omega}_1 + \dot{\sigma}_1) = -I_s \Omega^2 \omega_2 \sigma_3 + I_s \Omega^2 \omega_3 \sigma_2 - I_s \Omega^2 \mu \sigma_1$$

↓ $\div I_s \Omega^2$

$$\left\{ \begin{array}{l} (\dot{\omega}_1 + \dot{\sigma}_1) = -\omega_2 \sigma_3 + \omega_3 \sigma_2 - \mu \sigma_1 \\ \quad \downarrow \text{同理} \end{array} \right. \quad \text{✗}$$

$$\left\{ \begin{array}{l} (\dot{\omega}_2 + \dot{\sigma}_2) = -\omega_3 \sigma_1 + \omega_1 \sigma_3 - \mu \sigma_2 \\ (\dot{\omega}_3 + \dot{\sigma}_3) = -\omega_1 \sigma_2 + \omega_2 \sigma_1 - \mu \sigma_3 \end{array} \right. \quad \text{✗} \quad \left(\text{省略}^* \text{ 且 } (\dot{}) = \left(\frac{d}{dt^*} \right) \right)$$

$$\text{平衡解} \Rightarrow (\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3)_{\text{equil}} = (\Omega, 0, 0, 0, 0, 0)$$

$$\downarrow \omega_i^* = \frac{\omega_i}{\Omega}$$

$$(\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3)_{\text{equil}} = (1, 0, 0, 0, 0, 0) \quad \text{✗}$$

(9) 加入小擾動 $(\omega_1, \omega_2, \omega_3, \sigma_1, \sigma_2, \sigma_3) = (1 + \varepsilon \eta_1, \varepsilon \eta_2, \varepsilon \eta_3, \varepsilon \eta_4, \varepsilon \eta_5, \varepsilon \eta_6)$
 $(|\varepsilon| \ll 1)$

代入(4)式 $\Rightarrow$ (忽略 $O(\varepsilon^2)$ 以上)

$$\begin{cases} I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 + \mu \sigma_1 \\ I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_1 \omega_3 + \mu \sigma_2 \\ I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 + \mu \sigma_3 \end{cases}$$

$$\begin{cases} (\dot{\omega}_1 + \dot{\sigma}_1) = -\omega_2 \sigma_3 + \omega_3 \sigma_2 - \mu \sigma_1 \\ (\dot{\omega}_2 + \dot{\sigma}_2) = -\omega_3 \sigma_1 + \omega_1 \sigma_3 - \mu \sigma_2 \\ (\dot{\omega}_3 + \dot{\sigma}_3) = -\omega_1 \sigma_2 + \omega_2 \sigma_1 - \mu \sigma_3 \end{cases}$$

$$\Rightarrow \begin{cases} I_1 \dot{\eta}_1 = \mu \eta_4 \\ I_2 \dot{\eta}_2 = (I_3 - I_1) \eta_3 + \mu \eta_5 \\ I_3 \dot{\eta}_3 = (I_1 - I_2) \eta_2 + \mu \eta_6 \end{cases}$$

$$\Rightarrow \begin{cases} \dot{\eta}_1 + \dot{\eta}_4 = -\mu \eta_4 \\ \dot{\eta}_2 + \dot{\eta}_5 = -\mu \eta_5 + \eta_6 \\ \dot{\eta}_3 + \dot{\eta}_6 = -\eta_5 - \mu \eta_6 \end{cases}$$

分為2組

$$\begin{aligned} &\begin{matrix} \text{和} \\ \eta_1, \eta_4 \\ \text{有關} \end{matrix} \begin{cases} I_1 \dot{\eta}_1 = \mu \eta_4 \\ \dot{\eta}_1 + \dot{\eta}_4 = -\mu \eta_4 \end{cases} \\ &\begin{matrix} \text{和} \\ \eta_2, \eta_3, \\ \eta_5, \eta_6 \\ \text{有關} \end{matrix} \begin{cases} I_2 \dot{\eta}_2 = (I_3 - I_1) \eta_3 + \mu \eta_5 \\ I_3 \dot{\eta}_3 = (I_1 - I_2) \eta_2 + \mu \eta_6 \\ \dot{\eta}_2 + \dot{\eta}_5 = -\mu \eta_5 + \eta_6 \\ \dot{\eta}_3 + \dot{\eta}_6 = -\eta_5 - \mu \eta_6 \end{cases} \end{aligned}$$

(h) 令 $\vec{\eta}_L = \begin{bmatrix} \eta_1 \\ \eta_4 \end{bmatrix} \Rightarrow \begin{cases} I_1 \dot{\eta}_1 = \mu \eta_4 \\ \dot{\eta}_1 + \dot{\eta}_4 = -\mu \eta_4 \end{cases}$

$$\Rightarrow \underbrace{\begin{bmatrix} I_1 & 0 \\ 1 & 1 \end{bmatrix}}_{A} \ddot{\vec{\eta}}_L = \underbrace{\begin{bmatrix} 0 & \mu \\ 0 & -\mu \end{bmatrix}}_B \vec{\eta}_L$$

令 $\vec{\eta}_L = e^{st} \vec{c} \Rightarrow \vec{\eta}_L = s e^{st} \vec{c} = s \vec{\eta}_L$

$$\Rightarrow A\ddot{\eta}_L = sA\dot{\eta}_L = B\ddot{\eta}_L$$

$$\Rightarrow (sA-B)\ddot{\eta}_L = \vec{0} \quad (\ddot{\eta}_L \neq 0)$$

$$\Rightarrow \text{null}(sA-B) > 0$$

$$\Rightarrow \det(sA-B)=0 = \det \begin{bmatrix} sI_1 & -\mu \\ s & s+\mu \end{bmatrix}$$

$$= s(I_1 s + (I_1+1)\mu)$$

$$\Rightarrow s_1=0, s_2 = \frac{-\mu(I_1+1)}{I_1} \quad (\text{代回 } \ddot{\eta}_L = e^{st}\vec{c})$$

$$\Rightarrow \mu > 0 \text{ 且 } I_1 > 0 \Rightarrow s_2 < 0, s_1=0$$

$$\Rightarrow \text{所有 } s \text{ 实部} \leq 0 \text{ (含 } 0) \Rightarrow \text{中性稳定}$$

$$(ii) \text{ 令 } \ddot{\eta}_T = \begin{bmatrix} \eta_2 \\ \eta_3 \\ \eta_5 \\ \eta_6 \end{bmatrix} \Rightarrow \begin{cases} I_2 \dot{\eta}_2 = (I_3 - I_1)\eta_3 + \mu\eta_5 \\ I_3 \dot{\eta}_3 = (I_1 - I_2)\eta_2 + \mu\eta_6 \\ \dot{\eta}_2 + \dot{\eta}_5 = -\mu\eta_5 + \eta_6 \\ \dot{\eta}_3 + \dot{\eta}_6 = -\eta_5 - \mu\eta_6 \end{cases}$$

$$\Rightarrow \underbrace{\begin{bmatrix} I_2 & 0 & 0 & 0 \\ 0 & I_3 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}}_{A_{\times}} \ddot{\eta}_T = \underbrace{\begin{bmatrix} 0 & I_3 - I_1 & \mu & 0 \\ I_1 - I_2 & 0 & 0 & \mu \\ 0 & 0 & -\mu & 1 \\ 0 & 0 & -1 & -\mu \end{bmatrix}}_{B_{\times}} \ddot{\eta}_T$$

$$\Rightarrow \text{同上题 } \det(sA-B)=0 \text{ 解 } s$$

$$\Rightarrow \det(sA-B) = a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0 \Rightarrow \begin{cases} a_4 = I_2 I_3 \\ a_3 = \mu(2I_2 I_3 + I_2 + I_3) \\ a_2 = I_2 I_3 + (I_2 - I_1)(I_3 - I_1) + \mu^2(I_2 + 1)(I_3 + 1) \\ a_1 = \mu[2(I_2 - I_1)(I_3 - I_1) - (I_2 - I_1) - (I_3 - I_1)] \\ a_0 = (1 + \mu^2)(I_2 - I_1)(I_3 - I_1) \end{cases}$$

Routh-Hurwitz criteria $\Rightarrow$

$$s^4 \quad a_4 \quad a_2 \quad a_0$$

$$s^3 \quad a_3 \quad a_1$$

$$s^2 \quad \frac{-\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1 \quad a_0$$

$$s^1 \quad \frac{-\begin{vmatrix} a_3 & a_1 \\ b_1 & a_0 \end{vmatrix}}{b_1} = c_1$$

$$s^0 \quad a_0$$

已知 $I_1 > 0, I_2 > 0, I_3 > 0, \mu > 0$ $\Rightarrow$ $\begin{cases} a_4 = I_2 I_3 > 0 \\ a_3 = \mu(2I_2 I_3 + I_2 + I_3) > 0 \end{cases}$
耗散

情況 1 I_1 為中間車軸 $\begin{cases} I_2 < I_1 < I_3 \\ \text{或} \\ I_3 < I_1 < I_2 \end{cases}$

$\Rightarrow a_0 = (1+\mu^2)(I_2 - I_1)(I_3 - I_1) < 0 \Rightarrow$ Routh-Hurwitz 必失敗 $\Rightarrow$ 必有 s 在右半平面

$\Rightarrow$ 必有 $s, \text{ s.t. } \text{Re}(s) > 0$ ~~✗~~

$\Rightarrow$ 必不穩定 ~~✗~~

情況 2 I_1 為最大車軸 $\begin{cases} I_1 > I_2 \\ \text{且} \\ I_1 > I_3 \end{cases}$

by 邊長三角不等式 $I_1 < I_2 + I_3$

令 $\begin{cases} u = I_1 - I_2 > 0 \\ v = I_1 - I_3 > 0 \\ w = I_2 + I_3 - I_1 > 0 \end{cases}$

$\Rightarrow \begin{cases} I_1 = u + v + w \\ I_2 = v + w \\ I_3 = u + w \end{cases}$

$\Rightarrow \begin{cases} a_0 = uv(1+\mu^2) > 0 \\ a_1 = \mu(2uv + u + v) > 0 \\ a_2 = \mu^2 uv + \mu^2 uw + \mu^2 u + \mu^2 vw + \mu^2 v + \mu^2 w^2 + 2\mu^2 w + \mu^2 + 2uv + uw + vw + w^2 > 0 \\ a_3 = \mu(2uv + 2uw + u + 2vw + v + w^2 + 2w) > 0 \\ a_4 = (u + w)(v + w) > 0 \end{cases}$

$$b_1 = \frac{-\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = \frac{a_3 a_2 - a_4 a_1}{a_1} = \frac{\mu S_2(u, v, w, \mu)}{a_1}$$

$$S_2(u, v, w, \mu) \text{ 經計算係數皆為正} \Rightarrow S_2 > 0 \Rightarrow b_1 = \frac{\mu S_2}{a_1} > 0$$

$$c_1 = \frac{-\begin{vmatrix} a_3 & a_1 \\ b_1 & a_0 \end{vmatrix}}{b_1} = \frac{a_3 a_2 a_1 - a_4 a_1^2 - a_3^2 a_0}{a_3 a_2 - a_4 a_1} = \frac{\mu^2(u+v+w+1) S_3(u, v, w, \mu)}{\mu S_2(u, v, w, \mu)}$$

$$S_3(u, v, w, \mu) \text{ 經計算係數皆為正} \Rightarrow S_3 > 0 \Rightarrow c_1 = \frac{\mu(u+v+w+1) S_3}{S_2} > 0$$

by Routh-Hurwitz $\Rightarrow a_4, a_3, b_1, c_1, a_0$ 皆 $> 0 \Rightarrow$ 所有 s , s.t. $\text{Re}(s) < 0$ ~~✗~~
 $\Rightarrow I_1$ 為最大軸且 $\mu > 0$ 時, 系統為 asymptotically stable ~~✗~~

情況 3 I_1 為最小軸 $\begin{cases} I_1 < I_2 \\ \text{且} \\ I_1 < I_3 \end{cases}$

$$\text{令 } \begin{cases} I_1 = w \\ I_2 = w + u \\ I_3 = w + v \end{cases} \quad (u, v, w \text{ 皆} > 0)$$

$$\Rightarrow a_0 = uv(1 + \mu^2) > 0$$

$$c_1 = \frac{-\begin{vmatrix} a_3 & a_1 \\ b_1 & a_0 \end{vmatrix}}{b_1} = \frac{a_3 a_2 a_1 - a_4 a_1^2 - a_3^2 a_0}{a_3 a_2 - a_4 a_1} = \frac{-\mu^2(w+1) S_4(u, v, w, \mu)}{\mu S_5(u, v, w, \mu)}$$

$$S_3(u, v, w, \mu) \cdot S_5(u, v, w, \mu) \text{ 經計算係數皆為正} \Rightarrow c_1 < 0$$

$\Rightarrow$ Routh-Hurwitz 必失敗 $\Rightarrow$ 必有 s 在右半平面

$\Rightarrow$ 必有 s , s.t. $\text{Re}(s) > 0$ ~~✗~~

$\Rightarrow$ 必不穩定 ~~✗~~

~~✗~~ For 不穩定 sys ($\mu = 0$) $\Rightarrow \det(sA - B) = 0 = a_4 s^4 + a_2 s^2 + a_0 \Rightarrow$

$$\begin{cases} a_4 = I_2 I_3 > 0 \\ a_3 = 0 \\ a_2 = I_2 I_3 + (I_1 - I_2)(I_1 - I_3) \\ a_1 = 0 \\ a_0 = (I_2 - I_1)(I_3 - I_1) \end{cases}$$

Routh-Hurwitz $\Rightarrow$

$$\begin{array}{cccc}
 s^4 & a_4 & a_2 & a_0 \\
 s^3 & 0 & 0 & 0 \text{ 全零列} \Rightarrow \text{輔助力 eq.} = \frac{d}{ds}(\det(sA-B)) = 4a_4s^3 + 2a_2s = 0 \\
 s^3 & 4a_4 & 2a_2 & \\
 s^2 & \frac{a_2}{2} & a_0 & \\
 s^1 & \frac{a_2^2 - 4a_4a_0}{a_2/2} = b_1 & & \\
 s^0 & a_0 & &
 \end{array}$$

$\downarrow$
有一根 $s=0$

① I_1 最大或最小車由 $\Rightarrow a_2 > 0, a_0 > 0$, 且經計算 $b_1 > 0$

$\Rightarrow$ 但有一根 $s=0 \Rightarrow$ neutrally stable $\times$

② I_1 中間車由 $\Rightarrow a_0 < 0 \Rightarrow$ unstable $\times$ / 同中間車由定理

(j) 令 $I_2=2, I_3=1, \mu=0.1$

給 I_1 值, 解 $\det(sA-B)=0$ 的 4 個 s

(慣量 $\Rightarrow$ 解不等式 $I_2 - I_3 < I_1 < I_2 + I_3 \Rightarrow 1 < I_1 < 3$)

$$\Rightarrow \det(sA-B) = a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0 \Rightarrow \begin{cases} a_4 = I_2 I_3 \\ a_3 = \mu(2I_2 I_3 + I_2 + I_3) \\ a_2 = I_2 I_3 + (I_2 - I_1)(I_3 - I_1) + \mu^2(I_2 + 1)(I_3 + 1) \\ a_1 = \mu[2(I_2 - I_1)(I_3 - I_1) - (I_2 - I_1) - (I_3 - I_1)] \\ a_0 = (4\mu^2)(I_2 - I_1)(I_3 - I_1) \end{cases}$$

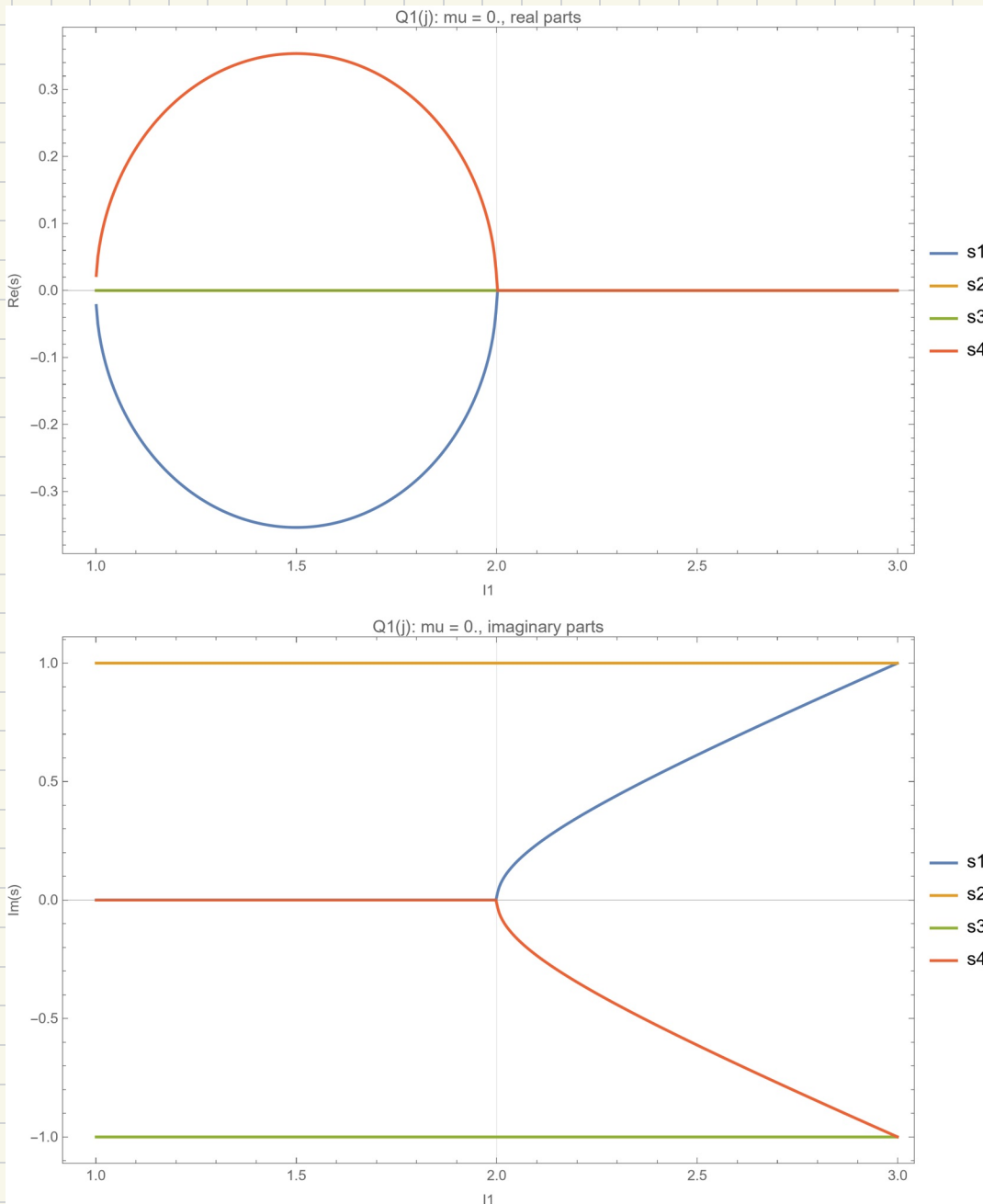
$$\Rightarrow \det(sA-B) = 2s^4 + 0.7s^3 + (I_1^2 - 3I_1 + 4.06)s^2 + 0.1(2I_1^2 - 4I_1 + 1)s + 1.01(I_1 - 1)(I_1 - 2)$$

① $1 < I_1 < 2 \Rightarrow I_1$ 為中間車由 $\Rightarrow$ 必有 $\operatorname{Re}(s) > 0$ (unstable)

② $2 < I_1 < 3 \Rightarrow I_1$ 為最大車由 $\Rightarrow \begin{cases} \mu > 0 \Rightarrow \text{asymptotically stable} \Rightarrow \forall s, \operatorname{Re}(s) < 0 \\ \mu = 0 \Rightarrow \text{neutrally stable} \Rightarrow \exists s, \operatorname{Re}(s) = 0 \end{cases}$

③ $I_1 = 2 \Rightarrow$ 有慣量重複 $\Rightarrow$ neutrally stable $\Rightarrow \exists s, \operatorname{Re}(s) = 0$

(1) $\mu=0$ (無耗能系統)

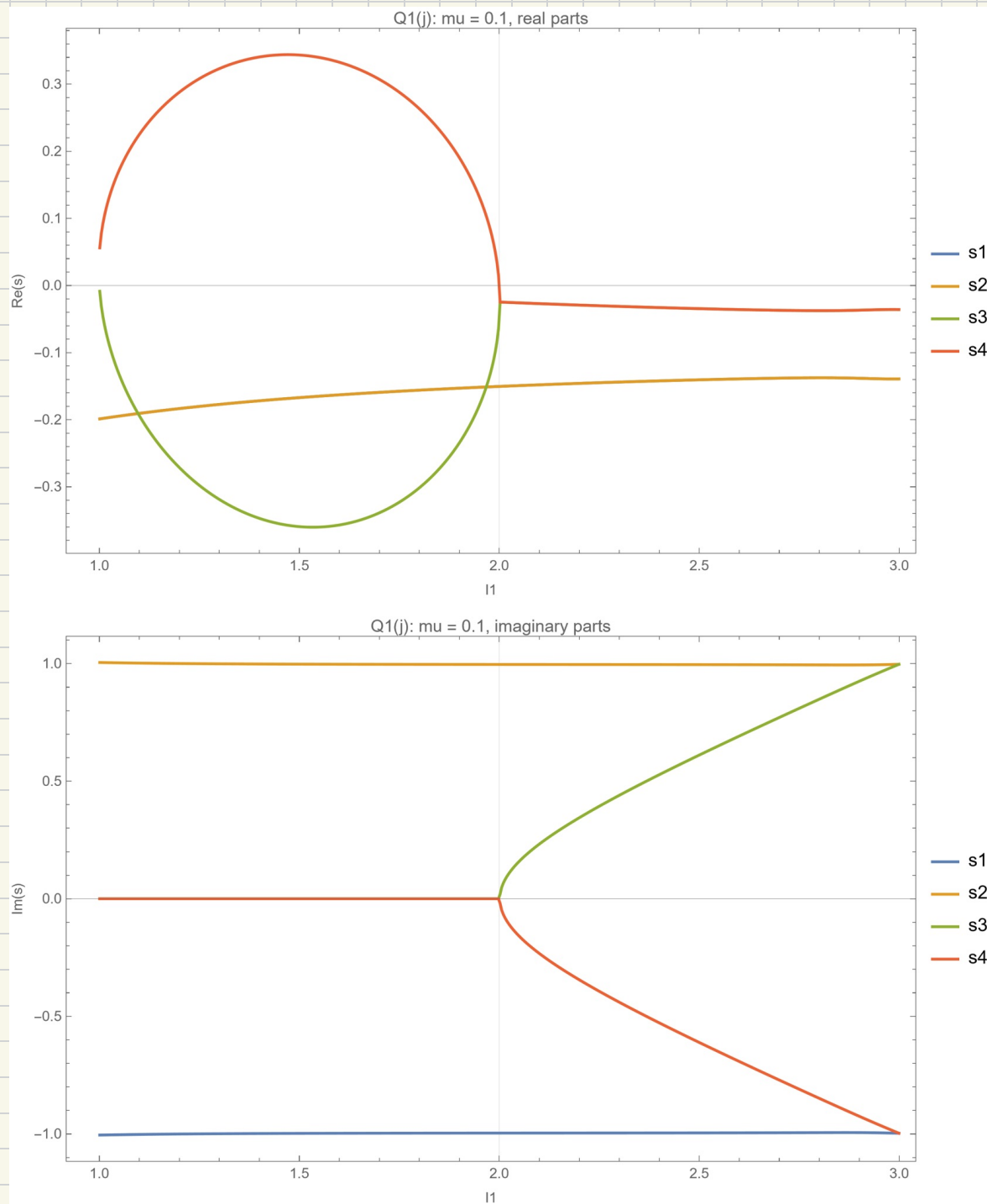


① $1 < I_1 < 2 \Rightarrow I_1$ 為中間軸 $\Rightarrow s$ 有實根對(-正-負) (s_1 及 s_4)
 $\Rightarrow \text{unstable}$

② $2 < I_1 < 3 \Rightarrow I_1$ 為最大軸 $\Rightarrow \forall s, \text{Re}(s) = 0 \Rightarrow \text{neutrally stable}$

③ $I_1 = 2 \Rightarrow$ 有慣量重複 $\Rightarrow \text{neutrally stable} \Rightarrow \forall s, s = 0 \Rightarrow \text{neutrally stable}$

(2) $\mu = 0.1 > 0$ (耗能系統)



① $1 < I_1 < 2 \Rightarrow I_1$ 為中間軸 $\Rightarrow \exists s, \text{s.t. } \text{Re}(s) > 0$ (s_2) (unstable) ✖

② $2 \leq I_1 < 3 \Rightarrow I_1$ 為最大軸 $\Rightarrow \mu > 0 \Rightarrow \forall s, \text{Re}(s) < 0 \Rightarrow \text{asymptotically stable}$ ✖

Q (b) - (c) - (d) - (j) 用 Mathematica code.

In[65]:=

```
ClearAll["Global`*"];
清除全部

SetOptions[{Plot, ListLinePlot, ParametricPlot},
指派選項 繪圖 點集的線繪圖 參數繪圖
Frame → True, ImageSize → 600, GridLines → Automatic, PlotRange → All];
邊框 真 影像尺寸 格線 自動 繪製範圍 全部

q1DefaultParams = <|"I" → {1., 2., 3.}, "Is" → 1. |>;
虛數單位

Q1AxisName[axis_Integer] := Switch[axis, 1, "minimum axis p1",
整數 切換
2, "intermediate axis p2", 3, "maximum axis p3", _, "unknown axis"];

Q1InitialState[axis_Integer, eps1_ : 0.02, eps2_ : -0.015] :=
整數
Module[{w = {0., 0., 0.}, others}, w[[axis]] = 1.;
模組
others = Complement[{1, 2, 3}, {axis}];
補集
w[[others[[1]]]] = eps1;
w[[others[[2]]]] = eps2;
Join[w, {0., 0., 0.}]];
連接

Q1RHS[muVal_?NumericQ, pars_ : q1DefaultParams][state_List] :=
數值量判定 清單
Module[{I1v, I2v, I3v, Isv, w1v, w2v, w3v, s1v, s2v, s3v, wdot, sphereRHS, sigdot},
模組
{I1v, I2v, I3v} = pars["I"];
虛數單位
Isv = pars["Is"];
{w1v, w2v, w3v, s1v, s2v, s3v} = state;
wdot = {(I2v - I3v) * w2v * w3v + muVal * s1v} / I1v, (I3v - I1v) * w3v * w1v +
muVal * s2v} / I2v, ((I1v - I2v) * w1v * w2v + muVal * s3v) / I3v};
sphereRHS = {(-muVal * s1v + Isv * w3v * s2v - Isv * w2v * s3v) / Isv,
(-Isv * w3v * s1v - muVal * s2v + Isv * w1v * s3v) / Isv,
(Isv * w2v * s1v - Isv * w1v * s2v - muVal * s3v) / Isv};
sigdot = sphereRHS - wdot;
Join[wdot, sigdot]];
連接

Q1Solve[axis_Integer, muVal_?NumericQ, tmax_ : 240, pars_ : q1DefaultParams] :=
整數 數值量判定
Module[{vars, rhs, init}, vars = {w1, w2, w3, sig1, sig2, sig3};
模組
rhs = Q1RHS[muVal, pars][Through[vars[t]]];
貫穿
init = Thread[Through[vars[0]] == Q1InitialState[axis]];
執行緒 貫穿
NDSolveValue[Join[Thread[{w1'[t], w2'[t], w3'[t], sig1'[t], sig2'[t], sig3'[t]} ==
常微分方程解... 連接 執行緒
```

```

    rhs], init], vars, {t, 0, tmax}, MaxSteps → Infinity,
    MaxStepFraction → 1 / 5000, AccuracyGoal → 11, PrecisionGoal → 11]];
    Q1HVector[state_List, pars_ : q1DefaultParams] :=
    Module[{Iv, Isv, wv, sigv}, Iv = pars["I"];
    Isv = pars["Is"];
    wv = state[[1 ;; 3]];
    sigv = state[[4 ;; 6]];
    Iv * wv + Isv * (wv + sigv)];
    Q1HNorm[state_List, pars_ : q1DefaultParams] := Norm[Q1HVector[state, pars]];
    Q1Energy[state_List, pars_ : q1DefaultParams] :=
    Module[{Iv, Isv, wv, sigv}, Iv = pars["I"];
    Isv = pars["Is"];
    wv = state[[1 ;; 3]];
    sigv = state[[4 ;; 6]];
    1 / 2 * Total[Iv * wv^2] + 1 / 2 * Isv * Total[(wv + sigv)^2]];
    Q1ScalarPlotRange[f_, {t0_, t1_}] :=
    Module[{v0, span}, v0 = Quiet@Check[N[f[t0]], 1.0];
    span = 0.05 * Max[Abs[v0], 1.0];
    {v0 - span, v0 + span}];
    Q1AxisData[axis_Integer, tmax_ : 240, pars_ : q1DefaultParams] := Module[
    {sol0, sold, state0, stated, h0, e0, hD, eD}, sol0 = Q1Solve[axis, 0., tmax, pars];
    sold = Q1Solve[axis, 0.1, tmax, pars];
    state0 = Function[{tt}, Through[sol0[tt]]];
    stated = Function[{tt}, Through[sold[tt]]];
    h0 = Function[{tt}, Q1HNorm[state0[tt], pars]];
    e0 = Function[{tt}, Q1Energy[state0[tt], pars]];
    hD = Function[{tt}, Q1HNorm[stateD[tt], pars]];
    eD = Function[{tt}, Q1Energy[stateD[tt], pars]]; <|"Axis" → axis, "Label" →
    Q1AxisName[axis], "tmax" → tmax, "pars" → pars, "sol0" → sol0, "sold" → sold,
    "state0" → state0, "stated" → stated, "h0" → h0, "e0" → e0, "hD" → hD, "eD" → eD|>;
    Q1RequiredFigureFromData[data_Association] :=

```

```

Module[{tmax, h0, e0, hD, eD, sol0, solD, omega0, omegaD, label}, tmax = data["tmax"];
  h0 = data["h0"];
  e0 = data["e0"];
  hD = data["hD"];
  eD = data["eD"];
  sol0 = data["sol0"];
  solD = data["solD"];
  label = data["Label"];

  omega0 = Through[sol0[[1 ;; 3]][t]];
  omegaD = Through[solD[[1 ;; 3]][t]];

  Column[{Style["Q1 required plots: initial spin mainly about " <> label, 15, Bold],
    Style["Required absolute plots for both mu=0 and mu=0.1:
      ||H|| (t), T(t), omega_i(t).", 11], Plot[h0[t], {t, 0, tmax},
        PlotLabel -> "mu = 0: ||H|| (t)", FrameLabel -> {"t", "||H||"},
        PlotRange -> Q1ScalarPlotRange[h0, {0, tmax}], ImageSize -> 640],
        Plot[e0[t], {t, 0, tmax}, PlotLabel -> "mu = 0: T(t)", FrameLabel -> {"t", "T"},
        PlotRange -> Q1ScalarPlotRange[e0, {0, tmax}], ImageSize -> 640],
        Plot[Evaluate[omega0], {t, 0, tmax}, PlotLabel -> "mu = 0: omega_i(t)",
        FrameLabel -> {"t", "omega_i"}, PlotLegends -> {"omega1", "omega2", "omega3"},
        ImageSize -> 640], Plot[hD[t], {t, 0, tmax}, PlotLabel -> "mu = 0.1: ||H|| (t)",
        FrameLabel -> {"t", "||H||"}, PlotRange -> Q1ScalarPlotRange[hD, {0, tmax}],
        ImageSize -> 640], Plot[eD[t], {t, 0, tmax}, PlotLabel -> "mu = 0.1: T(t)",
        FrameLabel -> {"t", "T"}, ImageSize -> 640], Plot[Evaluate[omegaD], {t, 0, tmax},
        PlotLabel -> "mu = 0.1: omega_i(t)", FrameLabel -> {"t", "omega_i"}, PlotLegends ->
        {"omega1", "omega2", "omega3"}, ImageSize -> 640]], Spacings -> 1.0]];

Q1RequiredFigure[axis_Integer] := Q1RequiredFigureFromData[Q1AxisData[axis]];

Q1TransverseA[I2v_, I3v_] := {{I2v, 0, 0, 0}, {0, I3v, 0, 0}, {1, 0, 1, 0}, {0, 1, 0, 1}};
Q1TransverseB[I1v_, I2v_, I3v_, muVal_] := {{0, I3v - I1v, muVal, 0},

```

```

    {I1v - I2v, 0, 0, muVal}, {0, 0, -muVal, 1}, {0, 0, -1, -muVal}};
Q1Eigenvalues[I1val_?NumericQ, muVal_?NumericQ, I2val_ : 2., I3val_ : 1.] :=
    [數值量判定] [數值量判定]
Module[{a, b}, a = N@Q1TransverseA[I2val, I3val];
    [模組] [數值化]
    b = N@Q1TransverseB[I1val, I2val, I3val, muVal];
    [數值化]
    Eigenvalues[{b, a}]];
    [特徵值]

Q1TrackedEigenvalues[muVal_?NumericQ,
    [數值量判定]
    I1min_ : 1.001, I1max_ : 2.999, npts_ : 500, I2val_ : 2., I3val_ : 1.] :=
Module[{xs, vals, nroot, tracked, perms, prev, current, bestPerm},
    [模組]
    xs = N@Subdivide[I1min, I1max, npts - 1];
    […] [細分]
    vals = (Q1Eigenvalues[#, muVal, I2val, I3val] &) /@ xs;
    nroot = Length[First[vals]];
    [長度] [第一個]
    tracked = ConstantArray[0. + 0. * I, {Length[xs], nroot}];
    [常數陣列] […] [長度]
    tracked[[1]] = SortBy[First[vals], {Re, Im}];
    [排序方式] [第一個] […] [虛部]
    perms = Permutations[Range[nroot]];
    [排列] [範圍]
    Do[prev = tracked[[k - 1]];
        [Do迴圈]
        current = vals[[k]];
        bestPerm = First@MinimalBy[perms, Total[Abs[current[[#]] - prev]^2] &];
        [第一個] [符合條件最小值] [總計] [絕對值]
        tracked[[k]] = current[[bestPerm]], {k, 2, Length[xs]};
        [長度]
    ];
    {xs, tracked}];

Q1EigenBranches[muVal_?NumericQ, part_ : Re, I1min_ : 1.001, I1max_ : 2.999,
    [數值量判定] [複數實部]
    npts_ : 500, I2val_ : 2., I3val_ : 1.] := Module[{xs, tracked, nroot},
    [模組]
    {xs, tracked} = Q1TrackedEigenvalues[muVal, I1min, I1max, npts, I2val, I3val];
    nroot = Length[First[tracked]];
    [長度] [第一個]
    Table[Transpose[{xs, part /@ tracked[[All, j]]}], {j, nroot}]];
    [表格] [轉置] [全部]

Q1EigenPlot[muVal_?NumericQ, I1min_ : 1.001, I1max_ : 2.999,
    [數值量判定]
    npts_ : 500, I2val_ : 2., I3val_ : 1.] := Module[{reData, imData, labels},
    [模組]
    reData = Q1EigenBranches[muVal, Re, I1min, I1max, npts, I2val, I3val];
    [複數實部]
    imData = Q1EigenBranches[muVal, Im, I1min, I1max, npts, I2val, I3val];
    [虛部]

```



```

labels = Table["s" <> ToString[j], {j, Length[reData] }];
Column[{ListLinePlot[reData, FrameLabel → {"I1", "Re(s)"},
PlotLabel → "Q1(j): mu = " <> ToString[muVal] <> ", real parts",
GridLines → {{2}, {0}}, PlotLegends → Placed[labels, Right], ImageSize → 640],
ListLinePlot[imData, FrameLabel → {"I1", "Im(s)"}, PlotLabel → "Q1(j): mu = " <>
ToString[muVal] <> ", imaginary parts", GridLines → {{2}, {0}},
PlotLegends → Placed[labels, Right], ImageSize → 640]], Spacings → 1.0]];

```

[illegible]

$$\vec{Y}_L = x \hat{i} + y \hat{j} = X \hat{i} + Y \hat{j}$$

$$\begin{cases} X = x \cos \phi - y \sin \phi \\ Y = x \sin \phi + y \cos \phi \end{cases} \quad (\phi = \omega t) \Rightarrow X^2 + Y^2 = x^2 + y^2$$

$$\begin{cases} \dot{X} = (\dot{x} - \dot{\phi}y)\cos\phi - (\dot{y} + \dot{\phi}x)\sin\phi \\ \dot{Y} = (\dot{x} - \dot{\phi}y)\sin\phi + (\dot{y} + \dot{\phi}x)\cos\phi \end{cases} \Rightarrow \dot{X}^2 + \dot{Y}^2 = (\dot{x} - \dot{\phi}y)^2 + (\dot{y} + \dot{\phi}x)^2 = (\dot{x} - \omega y)^2 + (\dot{y} + \omega x)^2 \quad (\dot{\phi} = \omega)$$

(1) Rayleigh's dissipation function =

$$\begin{aligned} R &= \frac{1}{2} c \dot{x}^2 + \frac{1}{2} c \dot{y}^2 + \frac{1}{2} c_b (b\dot{\beta}_1)^2 + \frac{1}{2} c_b (b\dot{\beta}_2)^2 \\ &= \frac{1}{2} c (\dot{x}^2 + \dot{y}^2) + \frac{1}{2} c_b (b\dot{\beta}_1)^2 + \frac{1}{2} c_b (b\dot{\beta}_2)^2 \\ &= \frac{1}{2} c [(\dot{x} - \omega y)^2 + (\dot{y} + \omega x)^2] + \frac{1}{2} c_b (b\dot{\beta}_1)^2 + \frac{1}{2} c_b (b\dot{\beta}_2)^2 \end{aligned}$$

(2) 簡單位能 $\Rightarrow V = \frac{1}{2} k (x^2 + y^2) = \frac{1}{2} k (x^2 + y^2)$

$$\begin{aligned}
 (a) \quad \text{令 } \begin{cases} A = (\dot{x} - \omega y) \\ B = (\dot{y} + \omega x) \end{cases} &\Rightarrow \vec{V}_C = \dot{\vec{r}}_C = \dot{x} \hat{i} + \dot{y} \hat{j} \\
 &= (A \cos \phi - B \sin \phi) \hat{i} + (A \sin \phi + B \cos \phi) \hat{j} \\
 &= \dot{x} \hat{i} + \dot{y} \hat{j} + \vec{\omega} \times (x \hat{i} + y \hat{j}) \\
 &= (\dot{x} - \omega y) \hat{i} + (\dot{y} + \omega x) \hat{j} \\
 &= A \hat{i} + B \hat{j}
 \end{aligned}$$

① for 圓盤 (質量 M 繞質心轉動慣量 I_D)

$$\begin{aligned}
 T_D &= \frac{1}{2} M \|\vec{V}_C\|^2 + \frac{1}{2} \vec{\omega} \cdot [\mathbf{I}] \vec{\omega} \\
 &= \frac{1}{2} M (A^2 + B^2) + \frac{1}{2} (\omega \hat{k}) \cdot (I_D \omega \hat{k}) \\
 &= \frac{1}{2} M (A^2 + B^2) + \frac{1}{2} I_D \omega^2
 \end{aligned}$$

② for 偏心質點 m_e

$$\begin{aligned}
 \vec{r}_e &= \vec{r}_C + e \hat{i} \\
 &\Downarrow \frac{d}{dt} \\
 \vec{v}_e &= \vec{v}_C + \vec{\omega} \times e \hat{i} \\
 &= A \hat{i} + B \hat{j} + e \omega \hat{j} \Rightarrow T_e = \frac{1}{2} m_e \|\vec{v}_e\|^2 \\
 &= A \hat{i} + (B + e\omega) \hat{j} \qquad \qquad \qquad = \frac{1}{2} m_e [A^2 + (B + e\omega)^2]
 \end{aligned}$$

③ for 第 i 顆滾珠

$$\begin{aligned}
 \vec{r}_i &= \vec{r}_C + (b \cos \beta_i) \hat{i} + (b \sin \beta_i) \hat{j} \\
 &\Downarrow \frac{d}{dt} \\
 \vec{v}_i &= \vec{v}_C - (b \dot{\beta}_i \sin \beta_i) \hat{i} + (b \dot{\beta}_i \cos \beta_i) \hat{j} + \vec{\omega} \times (b \cos \beta_i) \hat{i} + (b \sin \beta_i) \hat{j} \\
 &= A \hat{i} + B \hat{j} - (b \dot{\beta}_i \sin \beta_i) \hat{i} + (b \dot{\beta}_i \cos \beta_i) \hat{j} + (b \omega \cos \beta_i) \hat{j} - (b \omega \sin \beta_i) \hat{i} \\
 &= (A - b \dot{\beta}_i \sin \beta_i - b \omega \sin \beta_i) \hat{i} + (B + b \dot{\beta}_i \cos \beta_i + b \omega \cos \beta_i) \hat{j} \\
 &= (A - b \sin \beta_i (\dot{\beta}_i + \omega)) \hat{i} + (B + b \cos \beta_i (\dot{\beta}_i + \omega)) \hat{j}
 \end{aligned}$$

$$\Rightarrow T_{\tilde{x}} = \frac{1}{2} m_b \|\dot{\tilde{x}}\|^2 = \frac{1}{2} m_b \left[(A - b \sin \beta_{\tilde{x}} (\dot{\beta}_{\tilde{x}} + \omega))^2 + (B + b \cos \beta_{\tilde{x}} (\dot{\beta}_{\tilde{x}} + \omega))^2 \right]$$

$$\star T_{total} = T_D + T_e + T_1 + T_2$$

$$= \frac{1}{2} M \left[(\dot{x} - \omega y)^2 + (\dot{y} + \omega x)^2 \right] + \frac{1}{2} I_D \omega^2$$

$$+ \frac{1}{2} m_e \left[(\dot{x} - \omega y)^2 + (\dot{y} + \omega(x+e))^2 \right]$$

$$+ \frac{1}{2} m_b \left[(\dot{x} - \omega y - b \sin \beta_1 (\dot{\beta}_1 + \omega))^2 + (\dot{y} + \omega x + b \cos \beta_1 (\dot{\beta}_1 + \omega))^2 \right]$$

$$+ \frac{1}{2} m_b \left[(\dot{x} - \omega y - b \sin \beta_2 (\dot{\beta}_2 + \omega))^2 + (\dot{y} + \omega x + b \cos \beta_2 (\dot{\beta}_2 + \omega))^2 \right]$$

(b) 取廣義座標為 $\vec{q} = \begin{bmatrix} x \\ y \\ \beta_1 \\ \beta_2 \end{bmatrix}$ (ϕ 被定義 $\phi = \omega t \Rightarrow$ 不為廣義座標)

$$\text{彈性能} \Rightarrow V = \frac{1}{2} K (X^2 + Y^2) = \frac{1}{2} K (x^2 + y^2)$$

$$\text{Rayleigh's dissipation function} = (\dot{\phi} = \omega)$$

$$R = \frac{1}{2} c \dot{X}^2 + \frac{1}{2} c \dot{Y}^2 + \frac{1}{2} c_b (b \dot{\beta}_1)^2 + \frac{1}{2} c_b (b \dot{\beta}_2)^2$$

$$= \frac{1}{2} c (\dot{X}^2 + \dot{Y}^2) + \frac{1}{2} c_b (b \dot{\beta}_1)^2 + \frac{1}{2} c_b (b \dot{\beta}_2)^2$$

$$= \frac{1}{2} c \left[(\dot{x} - \omega y)^2 + (\dot{y} + \omega x)^2 \right] + \frac{1}{2} c_b (b \dot{\beta}_1)^2 + \frac{1}{2} c_b (b \dot{\beta}_2)^2$$

$$\Rightarrow \text{所需的力} = \begin{cases} \frac{\partial R}{\partial \dot{x}} = \frac{1}{2} c \cdot 2(\dot{x} - \omega y) = c(\dot{x} - \omega y) \\ \frac{\partial R}{\partial \dot{y}} = \frac{1}{2} c \cdot 2(\dot{y} + \omega x) = c(\dot{y} + \omega x) \\ \frac{\partial R}{\partial \dot{\beta}_1} = c_b b^2 \dot{\beta}_1 \\ \frac{\partial R}{\partial \dot{\beta}_2} = c_b b^2 \dot{\beta}_2 \end{cases}$$

(非保守廣義力)

⇒ Lagrange-Rayleigh eq. ⇒ $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} + \frac{\partial R}{\partial \dot{q}_j} = 0$ (除R外無非保守力)

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{q}_j} \right) - \frac{\partial I}{\partial q_j} + \frac{\partial V}{\partial \dot{q}_j} + \frac{\partial R}{\partial \dot{q}_j} = 0$$

$$\left(\text{其中 } \frac{d}{dt} = \sum_{j=1}^m \left(\dot{q}_j \frac{\partial}{\partial q_j} + \ddot{q}_j \frac{\partial}{\partial \dot{q}_j} \right) + \frac{\partial}{\partial t} \right)$$

```

In[93]:= ClearAll["Global`*"];
[清除全部]

q = {x[t], y[t], β1[t], β2[t]}; ⇒ 定義 q
qd = D[q, t];
[偏導]

qdd = D[q, {t, 2}];
[偏導]

qLabels = {"x", "y", "β1", "β2"};

params = {M, me, mb, ID, k, c, cb, b, e, ω};
$Assumptions = Element[params, Reals] &&
[預設假設] [屬於] [實數域]
M > 0 && me > 0 && mb > 0 && ID > 0 &&
k > 0 && c ≥ 0 && cb ≥ 0 && b > 0 && e ≥ 0;

simp[expr_] := Simplify[expr, $Assumptions];
[化簡] [預設假設]

```

→ $\vec{V}_C$

```

vC = {
D[x[t], t] - ω y[t],
[偏導]
D[y[t], t] + ω x[t]
[偏導]
};

```

→ $\vec{V}_E$

```

vE = {
D[x[t], t] - ω y[t],
[偏導]
D[y[t], t] + ω (x[t] + e)
[偏導]
};

```

→ $\vec{V}_i$

```

vB[ang_] := {
D[x[t], t] - ω y[t] - b Sin[ang] (D[ang, t] + ω),
[偏導] [正弦] [偏導]
D[y[t], t] + ω x[t] + b Cos[ang] (D[ang, t] + ω)
[偏導] [餘弦] [偏導]
};

```

⇓

計算 Lagrange-Rayleigh eq.

```

TD = 1/2 M (vC.vC) + 1/2 ID ω^2;
Te = 1/2 me (vE.vE);
Tb1 = 1/2 mb (vB[β1[t]].vB[β1[t]]); ⇒ 定義 T
Tb2 = 1/2 mb (vB[β2[t]].vB[β2[t]]);

```

⇒

```

T = simp[TD + Te + Tb1 + Tb2];

Rsupport = 1/2 c (vC.vC);
Rbeads = 1/2 cb (b D[β1[t], t])^2 +
[偏導]
1/2 cb (b D[β2[t], t])^2;
[偏導] ⇒ 定義 R

R = simp[Rsupport + Rbeads];

```

```

ELlhs[qi_] := simp[
D[D[T, D[qi, t]], t]
[偏導] [偏導]
- D[T, qi]
[偏導]
+ D[V, qi]
[偏導]
+ D[R, D[qi, t]]
[偏導] [偏導]
];

lhsRaw = ELlhs /@ q;

lhs = Collect[
[合併同類項]
Expand[lhsRaw],
[展開]
Join[qdd, qd],
[連接]
Simplify
[化簡]
];

eqs = Thread[lhs == 0];
[執行緒]

```

輸出

Out[119]=

q_i	Lagrange equation
x	$-\omega \left(b m b \cos(\beta_1(t)) + b m b \cos(\beta_2(t)) + e m e \right) + c y(t) - b m b \beta_1''(t) \sin(\beta_1(t)) - 2 b m b \omega \beta_1'(t) \cos(\beta_1(t)) - b m b \beta_1'(t)^2 \cos(\beta_1(t)) - b m b \beta_2''(t) \sin(\beta_2(t)) - 2 b m b \omega \beta_2'(t) \cos(\beta_2(t)) - b m b \beta_2'(t)^2 \cos(\beta_2(t)) + c x'(t) + x(t) \left(k - \omega^2 (M + 2 m b + m e) \right) + (M + 2 m b + m e) x''(t) - 2 \omega (M + 2 m b + m e) y'(t) = 0$
y	$b m b \beta_1''(t) \cos(\beta_1(t)) - 2 b m b \omega \beta_1'(t) \sin(\beta_1(t)) - b m b \beta_1'(t)^2 \sin(\beta_1(t)) - b m b \omega^2 (\sin(\beta_1(t)) + \sin(\beta_2(t))) + b m b \beta_2''(t) \cos(\beta_2(t)) - 2 b m b \omega \beta_2'(t) \sin(\beta_2(t)) - b m b \beta_2'(t)^2 \sin(\beta_2(t)) + c y(t) + c y'(t) + y(t) \left(k - \omega^2 (M + 2 m b + m e) \right) + 2 \omega (M + 2 m b + m e) x'(t) + (M + 2 m b + m e) y''(t) = 0$
β1	$b^2 c b \beta_1'(t) + b^2 m b \beta_1''(t) - b m b \sin(\beta_1(t)) x''(t) + 2 b m b \omega \cos(\beta_1(t)) x'(t) + b m b \omega^2 (x(t) \sin(\beta_1(t)) - y(t) \cos(\beta_1(t))) + b m b \cos(\beta_1(t)) y''(t) + 2 b m b \omega \sin(\beta_1(t)) y'(t) = 0$
β2	$b^2 c b \beta_2'(t) + b^2 m b \beta_2''(t) - b m b \sin(\beta_2(t)) x''(t) + 2 b m b \omega \cos(\beta_2(t)) x'(t) + b m b \omega^2 (x(t) \sin(\beta_2(t)) - y(t) \cos(\beta_2(t))) + b m b \cos(\beta_2(t)) y''(t) + 2 b m b \omega \sin(\beta_2(t)) y'(t) = 0$

整理如下 = 定義

$$\begin{cases} \text{總質量 } m = M + m_e + 2m_b \\ \text{形心 } C \text{ 的慣性加速度在 } x, y \text{ 座標上之分量} \end{cases} \begin{cases} a_x = \ddot{x} - 2\omega\dot{y} - \omega^2 x \\ a_y = \ddot{y} + 2\omega\dot{x} - \omega^2 y \end{cases}$$

$$\begin{pmatrix} \vec{a}_C = \vec{v}_C = (\ddot{x} - \omega\dot{y})\hat{i} + (\ddot{y} + \omega\dot{x})\hat{j} \\ + (\dot{x} - \omega y)\omega\hat{k} \times \hat{i} + (\dot{y} + \omega x)\omega\hat{k} \times \hat{j} \\ = \underbrace{(\ddot{x} - 2\omega\dot{y} - \omega^2 x)}_{a_x}\hat{i} + \underbrace{(\ddot{y} + 2\omega\dot{x} - \omega^2 y)}_{a_y}\hat{j} \end{pmatrix}$$

$$\textcircled{1} \text{ for } q_j = x \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{x}} \right) - \frac{\partial I}{\partial x} + \frac{\partial V}{\partial x} + \frac{\partial R}{\partial \dot{x}} = 0$$

$$\Rightarrow m a_x - m_e e \omega^2 - m_b b \sum_{i=1}^2 \left[\ddot{\beta}_i \sin \beta_i + (\omega + \dot{\beta}_i)^2 \cos \beta_i \right] + kx + C(\dot{x} - \omega y) = 0$$

$$\textcircled{2} \text{ for } q_j = y \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{y}} \right) - \frac{\partial I}{\partial y} + \frac{\partial V}{\partial y} + \frac{\partial R}{\partial \dot{y}} = 0$$

$$\Rightarrow m a_y + m_b b \sum_{i=1}^2 \left[\ddot{\beta}_i \cos \beta_i - (\omega + \dot{\beta}_i)^2 \sin \beta_i \right] + ky + C(\dot{y} + \omega x) = 0$$

$$\textcircled{3} \text{ for } q_j = \beta_1 \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{\beta}_1} \right) - \frac{\partial I}{\partial \beta_1} + \frac{\partial V}{\partial \beta_1} + \frac{\partial R}{\partial \dot{\beta}_1} = 0$$

$$\Rightarrow m_b b^2 \ddot{\beta}_1 + m_b b (a_y \cos \beta_1 - a_x \sin \beta_1) + C_b b^2 \dot{\beta}_1 = 0$$

$$\textcircled{3} \text{ for } q_j = \beta_2 \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{\beta}_2} \right) - \frac{\partial I}{\partial \beta_2} + \frac{\partial V}{\partial \beta_2} + \frac{\partial R}{\partial \dot{\beta}_2} = 0$$

$$\Rightarrow m_b b^2 \ddot{\beta}_2 + m_b b (a_y \cos \beta_2 - a_x \sin \beta_2) + C_b b^2 \dot{\beta}_2 = 0$$

上4式對 $\vec{q} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{\beta}_1 \\ \ddot{\beta}_2 \end{bmatrix}$ 為線性 $\Rightarrow$ 可寫成矩陣型式 $\Rightarrow [M(q)] \ddot{\vec{q}} = \vec{F}(\vec{q}, \dot{\vec{q}}) \Rightarrow$ 可用以數值積分

$$\underbrace{\begin{bmatrix} m & 0 & -m_b b \sin \beta_1 & -m_b b \sin \beta_2 \\ 0 & m & m_b b \cos \beta_1 & m_b b \cos \beta_2 \\ -m_b b \sin \beta_1 & m_b b \cos \beta_1 & m_b b^2 & 0 \\ -m_b b \sin \beta_2 & m_b b \cos \beta_2 & 0 & m_b b^2 \end{bmatrix}}_{[M(q)]} \ddot{\vec{q}} = \vec{F}(\vec{q}, \dot{\vec{q}}) = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$

$$\begin{cases} F_1 = 2m\omega\dot{y} + m\omega^2 x + m_e e \omega^2 + m_b b \sum_{i=1}^2 (\omega + \dot{\beta}_i)^2 \cos \beta_i - kx - C(\dot{x} - \omega y) \\ F_2 = -2m\omega\dot{x} + m\omega^2 y + m_b b \sum_{i=1}^2 (\omega + \dot{\beta}_i)^2 \sin \beta_i - ky - C(\dot{y} + \omega x) \\ F_3 = -m_b b [(2\omega\dot{x} - \omega^2 y) \cos \beta_1 + (2\omega\dot{y} + \omega^2 x) \sin \beta_1] - C_b b^2 \dot{\beta}_1 \\ F_4 = -m_b b [(2\omega\dot{x} - \omega^2 y) \cos \beta_2 + (2\omega\dot{y} + \omega^2 x) \sin \beta_2] - C_b b^2 \dot{\beta}_2 \end{cases}$$

~~✗~~

(C) 平衡了解 $\Rightarrow$ i.e. $\frac{d}{dt} = 0 \Rightarrow \dot{x} = \dot{y} = \dot{\beta}_1 = \dot{\beta}_2 = \ddot{x} = \ddot{y} = \ddot{\beta}_1 = \ddot{\beta}_2 = 0$

此時 $\vec{V}_c = a_x \hat{i} + a_y \hat{j}$, $\begin{cases} a_x = -\omega^2 x \\ a_y = -\omega^2 y \end{cases} \quad \underline{A} \begin{cases} A = -\omega y \\ B = \omega x \end{cases}$

代入(b)之 Lagrange 運動方程

① for $q_j = x \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{x}} \right) - \frac{\partial I}{\partial x} + \frac{\partial V}{\partial x} + \frac{\partial R}{\partial \dot{x}} = 0$

$$\Rightarrow (k - m\omega^2)x - C\omega y = \omega^2 [m_e e + m_b b (\cos \beta_1 + \cos \beta_2)]$$

② for $q_j = y \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{y}} \right) - \frac{\partial I}{\partial y} + \frac{\partial V}{\partial y} + \frac{\partial R}{\partial \dot{y}} = 0$

$$\Rightarrow C\omega x + (k - m\omega^2)y = \omega^2 m_b b (\sin \beta_1 + \sin \beta_2)$$

③ for $q_j = \beta_1 \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{\beta}_1} \right) - \frac{\partial I}{\partial \beta_1} + \frac{\partial V}{\partial \beta_1} + \frac{\partial R}{\partial \dot{\beta}_1} = 0$

$$\Rightarrow x \sin \beta_1 - y \cos \beta_1 = 0$$

③ for $q_j = \beta_2 \Rightarrow \frac{d}{dt} \left(\frac{\partial I}{\partial \dot{\beta}_2} \right) - \frac{\partial I}{\partial \beta_2} + \frac{\partial V}{\partial \beta_2} + \frac{\partial R}{\partial \dot{\beta}_2} = 0$

$$\Rightarrow x \sin \beta_2 - y \cos \beta_2 = 0$$

令 $\lambda = k - m\omega^2$ (徑向剛度, 彈簧 k 扣離心力慣性)

$\delta = C\omega$ (固定座標阻尼切向效果) $\xrightarrow{\text{① ②}} \begin{bmatrix} \lambda & -\delta \\ \delta & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \omega^2 \begin{bmatrix} F + b(\cos \beta_1 + \cos \beta_2) \\ b(\sin \beta_1 + \sin \beta_2) \end{bmatrix}$

$F = m_e e$ (偏心質量不平衡量)

$b = m_b b$ (單滾珠不平衡量)

$$\text{令 } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad (\text{換成極座標}) \quad \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1}(\frac{y}{x}) \end{cases}$$

$$(1) r=0 \text{ (震幅平衡)} \Rightarrow \begin{cases} x=0 \\ y=0 \end{cases} \quad \text{③④式自動滿足}$$

$$\begin{bmatrix} \lambda & -\delta \\ \delta & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \omega^2 \begin{bmatrix} F + b(\cos \beta_1 + \cos \beta_2) \\ b(\sin \beta_1 + \sin \beta_2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} F + b(\cos \beta_1 + \cos \beta_2) = 0 \\ b(\sin \beta_1 + \sin \beta_2) = 0 \end{cases} \Rightarrow F \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} \cos \beta_1 \\ \sin \beta_1 \end{bmatrix} + b \begin{bmatrix} \cos \beta_2 \\ \sin \beta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

偏心質點造成之
旋轉不平衡量
(在+x方向)

兩滾珠造成之旋轉不平衡量各為b
各在方向 $\begin{bmatrix} \cos \beta_1 \\ \sin \beta_1 \end{bmatrix}$ 及 $\begin{bmatrix} \cos \beta_2 \\ \sin \beta_2 \end{bmatrix}$ 上

↓
若同方向且皆在+x $\Rightarrow \beta_1 = \beta_2 = 0$

$\Rightarrow$ b在+x向有最大不平衡量 $= 2b$

$\Rightarrow$ 兩顆滾珠可提供之合成力矩須抵消偏心質量旋轉不平衡量F

$\Rightarrow$ 兩顆滾珠最大可提供之合成力矩為 $2b = 2m_b b$

$\Rightarrow$ 因此震幅平衡存在之充要條件為 $F \leq 2b \Rightarrow \text{i.e. } m_e e \leq 2m_b b$
(β_1, β_2 有解)

$$(i) \text{ if } m_e e < 2m_b b \Rightarrow \text{令 } \begin{cases} \beta_1 = \pi + \alpha \\ \beta_2 = \pi - \alpha \end{cases}$$

$$b(\sin \beta_1 + \sin \beta_2) = 0 \text{ 自動成立}$$

$$F + b(\cos \beta_1 + \cos \beta_2) = 0 = F - 2b \cos \alpha \Rightarrow \alpha = \cos^{-1}\left(\frac{F}{2b}\right) = \cos^{-1}\left(\frac{m_e e}{2m_b b}\right)$$

(ii) if $m_{ee} = 2m_{bb} \Rightarrow \alpha = 0 \Rightarrow \beta_1 = \beta_2 = \pi$ ✖

(iii) if $m_{ee} > 2m_{bb} \Rightarrow$ 零振幅平衡不存在 ✖

(2) $\gamma > 0$ ($\Rightarrow$ 非零振幅平衡)

$$\textcircled{3} \Rightarrow x \sin \beta_i - y \cos \beta_i = 0 = r \cos \theta \sin \beta_i - r \sin \theta \cos \beta_i = r \sin(\beta_i - \theta) = 0$$

$$\Rightarrow \sin(\beta_i - \theta) = 0 \Rightarrow \boxed{\beta_i = \theta \vee \theta + \pi} \quad (\beta_i = \theta + n\pi \quad (n \in \mathbb{Z}))$$

每顆滾珠須位於形心位移方向或其反方向

$$\text{令 } \varepsilon_i = \begin{cases} +1, & \text{if } \beta_i = \theta \\ -1, & \text{if } \beta_i = \theta + \pi \end{cases}, \text{ 且 } E = \varepsilon_1 + \varepsilon_2 \quad (E \in \{-2, 0, +2\})$$

$$(\text{位於 } \theta \text{ 方向} \Rightarrow \text{定義 } \begin{cases} \text{徑向 } \hat{e}_r = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \\ \text{切向 } \hat{e}_\theta = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} \end{cases})$$

$$\text{一顆滾珠不平衡向量爲 } g \begin{bmatrix} \cos \beta_i \\ \sin \beta_i \end{bmatrix} \text{ 且 } \beta_i = \theta \vee \theta + \pi \Rightarrow \begin{cases} \beta_i = \theta \Rightarrow \text{不平衡量} = g \hat{e}_r \\ \beta_i = \theta + \pi \Rightarrow \text{不平衡量} = -g \hat{e}_r \end{cases}$$

$$\Rightarrow \text{此時兩球合成方向不平衡量爲 } gE \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

$$\textcircled{1} \Rightarrow \begin{bmatrix} \lambda & -\delta \\ \delta & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \omega^2 \begin{bmatrix} F + g(\cos \beta_1 + \cos \beta_2) \\ g(\sin \beta_1 + \sin \beta_2) \end{bmatrix}$$

$$= \omega^2 \left(F \begin{bmatrix} 1 \\ 0 \end{bmatrix} + g \begin{bmatrix} \cos \beta_1 \\ \sin \beta_1 \end{bmatrix} + g \begin{bmatrix} \cos \beta_2 \\ \sin \beta_2 \end{bmatrix} \right)$$

$$= \omega^2 \left(F \begin{bmatrix} 1 \\ 0 \end{bmatrix} + gE \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \right)$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = r \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = r \hat{e}_r$$

$$\Rightarrow r \begin{bmatrix} \lambda & -\delta \\ \delta & \lambda \end{bmatrix} \hat{e}_r = r \begin{bmatrix} \lambda \cos \theta - \delta \sin \theta \\ \delta \cos \theta + \lambda \sin \theta \end{bmatrix} = \lambda r \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} + \delta r \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} \\ = \lambda r \hat{e}_r + \delta r \hat{e}_\theta$$

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \cos \theta \hat{e}_r - \sin \theta \hat{e}_\theta$$

$$\Rightarrow \omega^2 \left(F \begin{bmatrix} 1 \\ 0 \end{bmatrix} + h E \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \right) = \omega^2 F \cos \theta \hat{e}_r - \omega^2 F \sin \theta \hat{e}_\theta + h E \hat{e}_r \\ = \omega^2 (F \cos \theta + h E) \hat{e}_r - \omega^2 F \sin \theta \hat{e}_\theta \\ = \lambda r \hat{e}_r + \delta r \hat{e}_\theta$$

$$\Rightarrow \begin{cases} \lambda r = \omega^2 (F \cos \theta + h E) & (\text{徑向}) \\ \delta r = -\omega^2 F \sin \theta & (\text{切向}) \end{cases}$$

$$\Rightarrow \begin{aligned} & (\lambda r - \omega^2 h E = \omega^2 F \cos \theta)^2 \\ & (\delta r = -\omega^2 F \sin \theta)^2 \end{aligned} \quad (+) \\ \hline (\delta r)^2 + (\lambda r - \omega^2 h E)^2 = (\omega^2 F)^2$$

$$\Rightarrow (\lambda^2 + \delta^2) r^2 - 2\lambda \omega^2 h E r + \omega^4 (h^2 E^2 - F^2) = 0$$

$$\Rightarrow r = \frac{\omega^2 [\lambda h E \pm \sqrt{(\lambda^2 + \delta^2) F^2 - \delta^2 h^2 E^2}]}{\lambda^2 + \delta^2}$$

$$\begin{cases} \lambda = k - m\omega^2 & (\text{徑向剛度, 彈簧 } k \text{ 扣齊心慣性}) \\ \delta = c\omega & (\text{固定座標阻尼切向效果}) \\ F = me & (\text{偏心質點不平衡量}) \\ h = mb & (\text{單滾珠不平衡量}) \\ E = \varepsilon_1 + \varepsilon_2 \in \{-2, 0, +2\} \end{cases}$$

✱ 存在條件為 $(\lambda^2 + \delta^2) F^2 - \delta^2 h^2 E^2 > 0$ 且 $r > 0$ ✱

$$\theta = \sin^{-1}\left(-\frac{\delta r}{\omega^2 F}\right) \text{ 或 } \cos^{-1}\left(\frac{\lambda r - \omega^2 G E}{\omega^2 F}\right)$$

~~✗~~

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

(d) natural frequency $\omega_n = \sqrt{\frac{k}{m}}$

取 $M=1, m_e=0.05, m_b=0.05, e=0.2, b=1, c=0.05, c_b=0.05$

且 $m = M + m_e + 2m_b = 1.15$

取 $k_f = 1.15 \Rightarrow \omega_n = \sqrt{\frac{k}{m}} = 1$

$$\begin{cases} F = m_e e = 0.01 \\ G = m_b b = 0.05 \Rightarrow 2G = 0.1 \end{cases}$$

(1) for $r=0$ 零振幅平衡 $\begin{cases} x=0 \\ y=0 \end{cases}$

$m_e e < 2m_b b \Rightarrow \text{令 } \begin{cases} \beta_1 = \pi + \alpha \\ \beta_2 = \pi - \alpha \end{cases}$

$G(\sin \beta_1 + \sin \beta_2) = 0$ 自動成立

$$F + G(\cos \beta_1 + \cos \beta_2) = 0 = F - 2G \cos \alpha \Rightarrow \alpha = \cos^{-1}\left(\frac{F}{2G}\right) = \cos^{-1}\left(\frac{m_e e}{2m_b b}\right)$$

$$= \cos^{-1}(0.1) \approx 1.4106 \text{ (rad)}$$

$$\Rightarrow \begin{cases} \beta_1 = \pi + 1.4106 \\ \beta_2 = \pi - 1.4106 \end{cases}$$

※ 測試 零振幅平衡 是否具有吸引力 $\Rightarrow$ 取初始條件 = (零振幅平衡 + 小擾動)
(加入擾動後仍回到 $r=0$)

I.C. $\Rightarrow \begin{cases} x(0) = 10^{-3} \\ y(0) = 0 \\ \beta_1(0) = \pi + \alpha + 0.05 \\ \beta_2(0) = \pi - \alpha - 0.05 \end{cases}$

且 $\dot{x}(0) = \dot{y}(0) = \dot{\beta}_1(0) = \dot{\beta}_2(0) = 0$

意義：測試 I.C. 幾乎為理想平衡狀態，加入擾動後是否能回穩 ($r=0$)

① 低臨界轉速 (取 $\omega = 0.8 < \omega_n = 1$)

$$\Rightarrow \begin{cases} \lambda = k - m\omega^2 = 0.414 \\ \delta = c\omega = 0.040 \end{cases}$$

by (c) 非零平衡須滿足 $(\delta r)^2 + (\lambda r - \omega^2 GE)^2 = (\omega^2 F)^2$

$\Rightarrow$ 平衡解可為以下

型態	E	r	θ	解讀
零振幅	-	0	未定	兩球對稱分布，存在但不一定穩定
非零	0	0.01539	-0.0963	一球沿 θ ，一球反向
非零	2	0.14894	-1.9448	兩球沿 θ ，候選解
非零	2	0.15738	-1.3895	兩球沿 θ ，數值吸引解

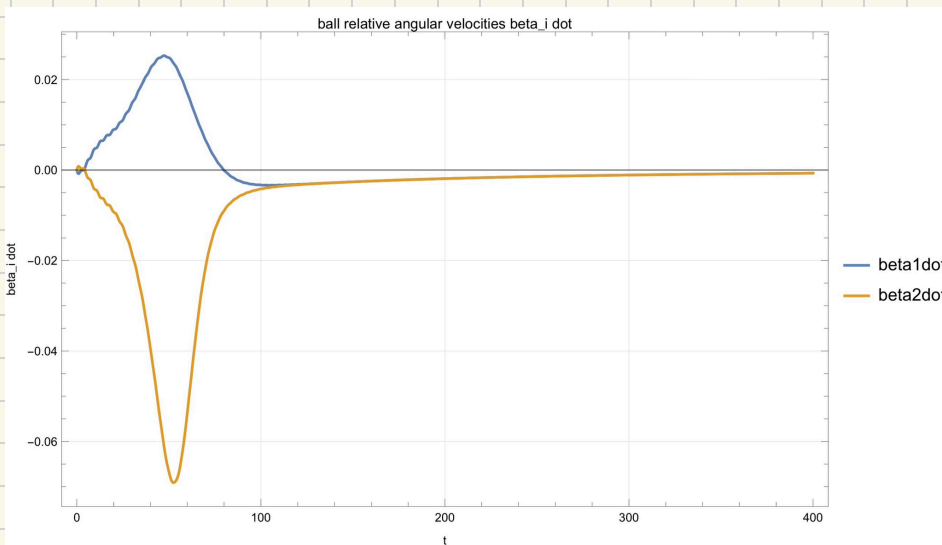
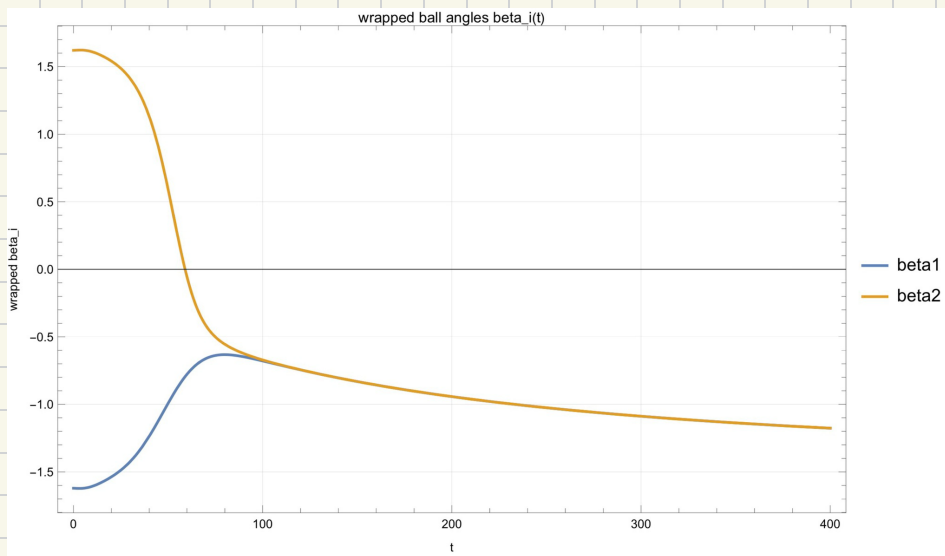
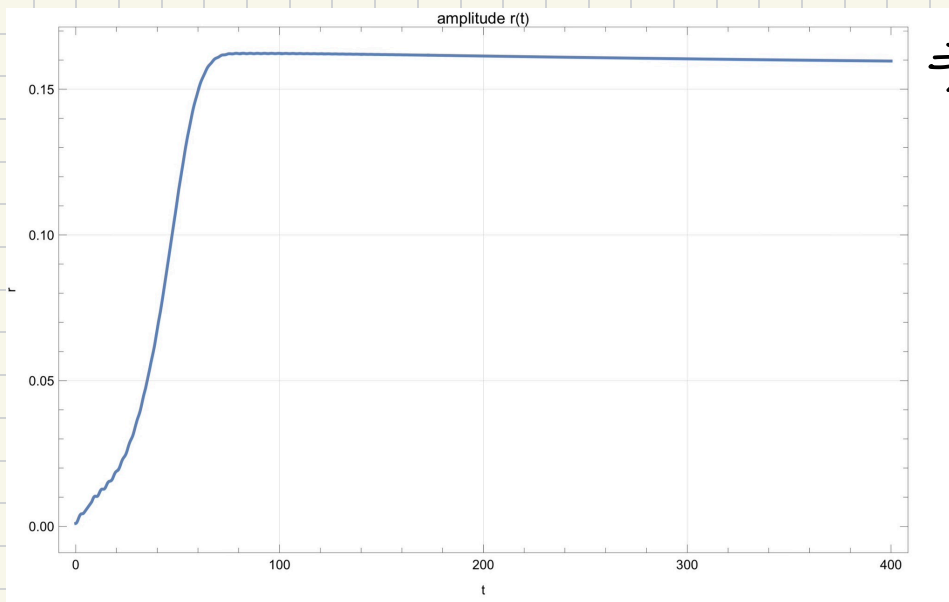
為了預測長時間行為 $\Rightarrow$ 對各平衡解點做數值線性化

結果： $\begin{cases} \text{零振幅平衡系統最大特徵值實部為正} \Rightarrow \text{存在但不為吸引態} \\ \text{其一非零振幅平衡系統最大特徵值實部為負} \Rightarrow \text{此參數下之局部吸引解} \end{cases}$

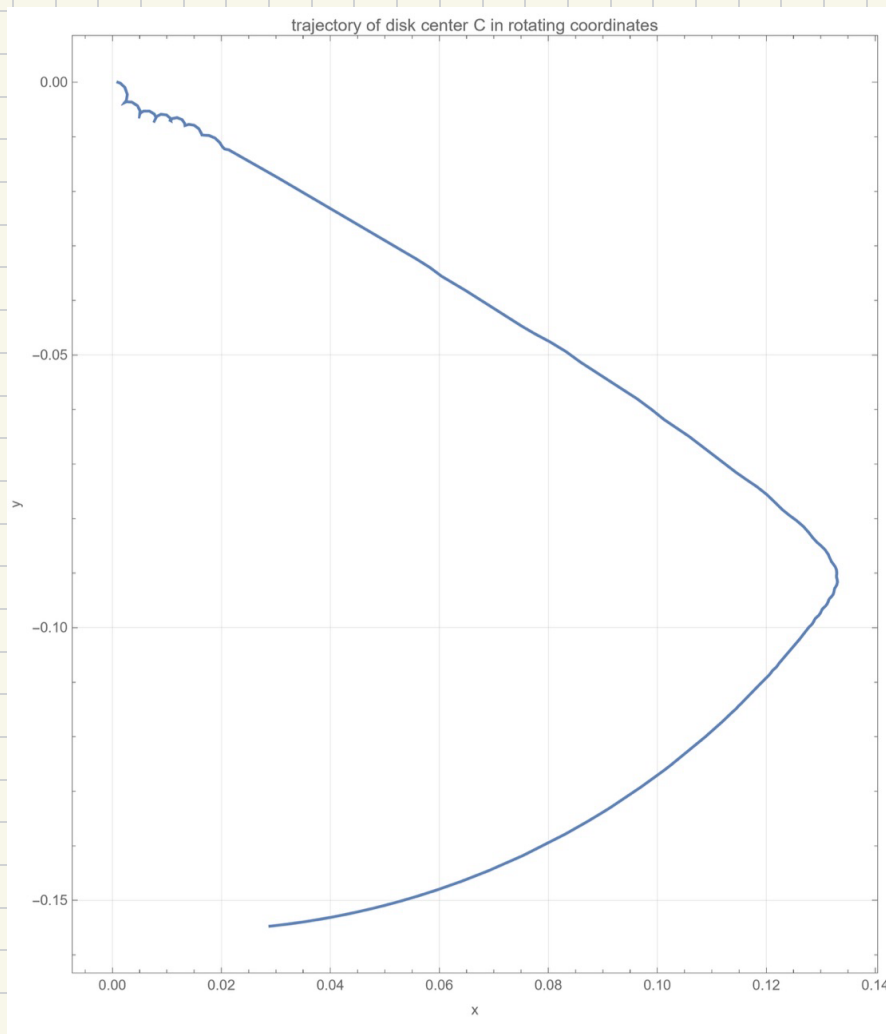
型態	E	r	θ	最大特徵值實部
零振幅	-	0	-	+0.0654
非零	0	0.01539	-0.0963	+0.0669
非零	2	0.14894	-1.9448	+0.00233
非零	2	0.15738	-1.3895	-0.00239

預測 = 低臨界轉速 ($\omega = 0.8 < \omega_n = 1$) 時，從零振幅平衡加入小擾動 $\Rightarrow$ 軌跡亦離開零振幅平衡，並最後收斂至 $\begin{cases} r \approx 0.1574 \\ \theta \approx -1.39 \end{cases}$

用 mathematica 模擬：



开衫心 C 軌跡圖



C 最终停在约 $(x, y) \approx (0.0284, -0.1548)$
 $\Rightarrow r \approx 0.1574$ (吻合)
~~X~~

結論: 零振幅平衡存在, 但不吸引, 最终趋近稳定的非零振幅平衡
 (不稳定) ~~X~~

① 過臨界轉速 (取 $\omega = 1.2 > \omega_n = 1$)

$$\Rightarrow \begin{cases} \lambda = k - m\omega^2 = -0.506 \\ \delta = c\omega = 0.060 \end{cases}$$

$\Rightarrow m_e c < 2m_b b \Rightarrow$ 故零振幅平衡存在, 但未必穩定

可能平衡解如下:

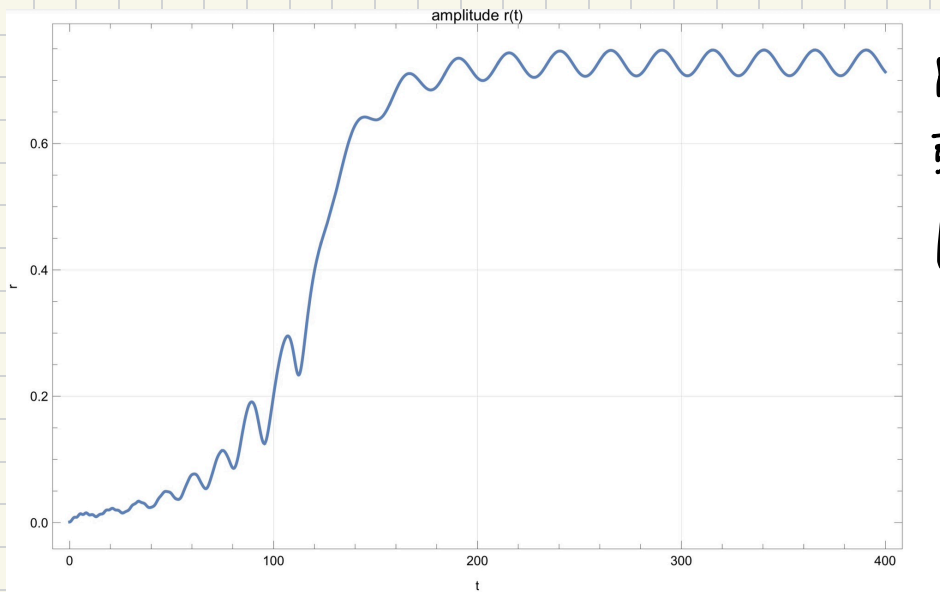
型態	E	r	θ	解讀
零振幅	-	0	未定	因 $m_e c < 2m_b b$ 而存在
非零	-2	無	-	判別式小於零
非零	0	0.02826	-3.0236	一球沿 θ , 一球反向
非零	2	無	-	判別式小於零

對各平衡點做數值線性化：

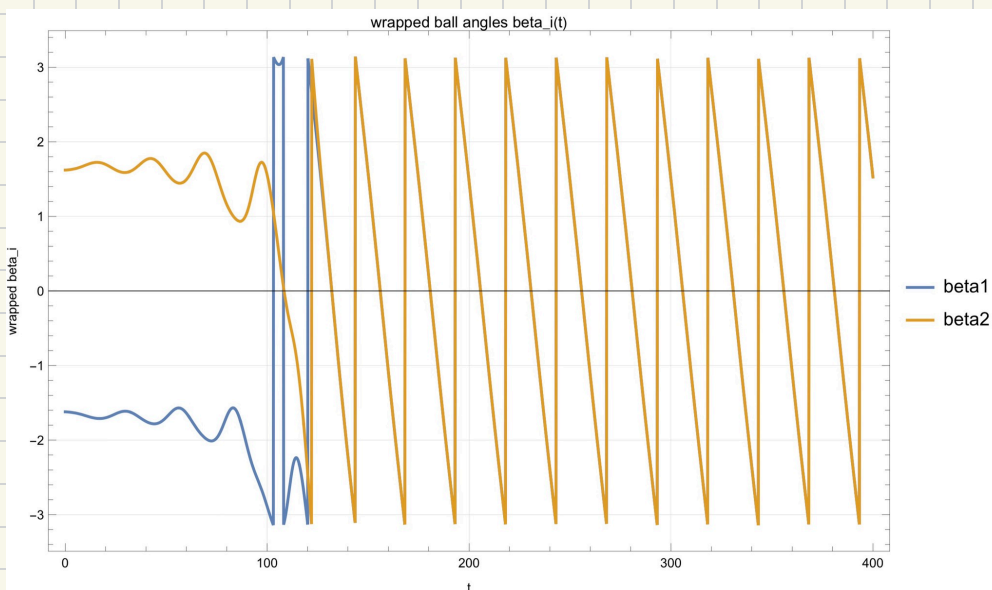
型態	ε	r	θ	最大特徵值實部
零振幅	-	0	-	+0.0311
非零	0	0.02826	-3.0236	+0.0299

⇒ 實部皆 > 0 ⇒ 零/非零振幅平衡皆不吸引 (穩定)

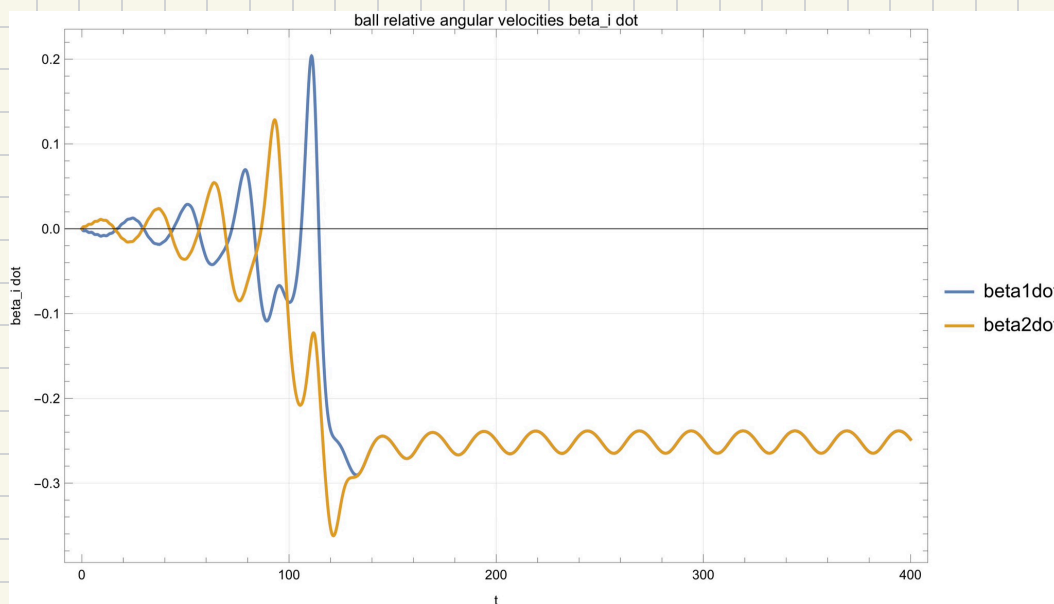
預測：過臨界轉速 ($\omega = 1.2 > \omega_n = 1$) 時，從零振幅平衡加入小擾動 ⇒ 軌跡離開零振幅平衡且不收斂
⇒ 進入非平衡長時間運動 (如：類週期運動)



r 從 0 開始成長，後段不趨近零或任何數值，而是在 0.7 附近這週期性振盪

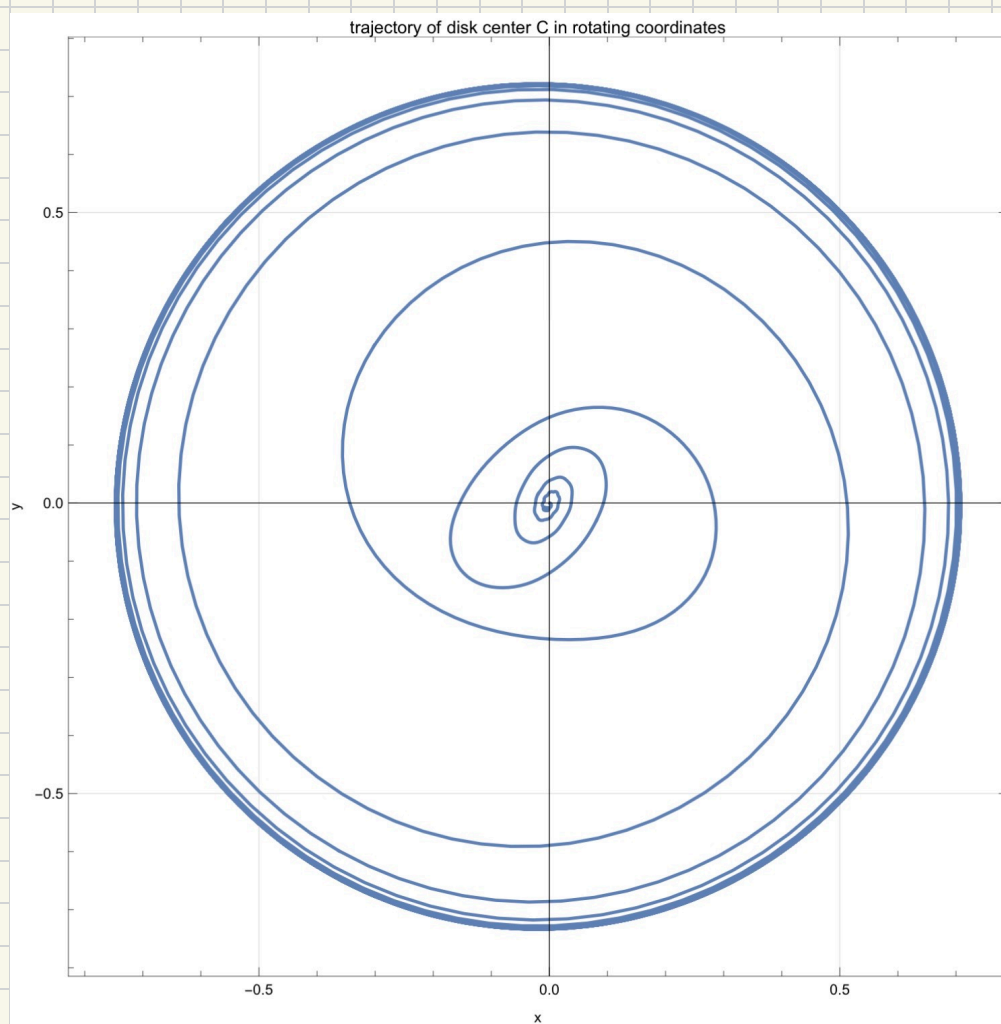


滾珠角度 β_1, β_2 週期性齒狀跳躍



β_1, β_2 後段在負值小幅振盪
 $\Rightarrow$ 表示滾珠不固定在某一角度上, 持續相對圓盤滾動

形心 C 軌跡圖



$\begin{cases} \omega < \omega_n \Rightarrow C \text{ 趨向固定點} \\ \omega > \omega_n \Rightarrow C \text{ 形成一圈一圈} \end{cases}$
 近似閉合的軌道
 $\Rightarrow$ 非平衡長時間週期運動

結論: 平衡態平衡解存在, 但不為吸引態(不穩定), 系統進入非平衡長時間週期運動(呈近似週期軌道)

~~✗~~

★ $2m_b b$ 和 $m_e c$ 的比值關係 = (同(c)子題)

定義平衡能力比值 $\rho = \frac{2m_b b}{m_e c} = \frac{\text{兩滾珠最大合成質量矩}}{\text{偏心質量矩}}$

① 若 $\rho < 1 \Rightarrow$ 兩滾珠最大合成質量矩 (皆於偏心質量正對面) 仍無法抵消
偏心質量矩 $\Rightarrow$ 零振幅平衡不存在 ✖

② $\rho = 1 \Rightarrow$ 零振幅平衡剛好存在, 但必須 $\beta_1 = \beta_2 = \pi$ ✖

③ $\rho > 1 \Rightarrow$ 零振幅平衡存在, 兩滾珠可對稱分佈於偏心質量反方向附近

$$\Rightarrow \begin{cases} \beta_1 = \pi + \alpha \\ \beta_2 = \pi - \alpha \end{cases} \left(\alpha = \cos^{-1}\left(\frac{1}{\rho}\right) \right) \text{ ✖}$$

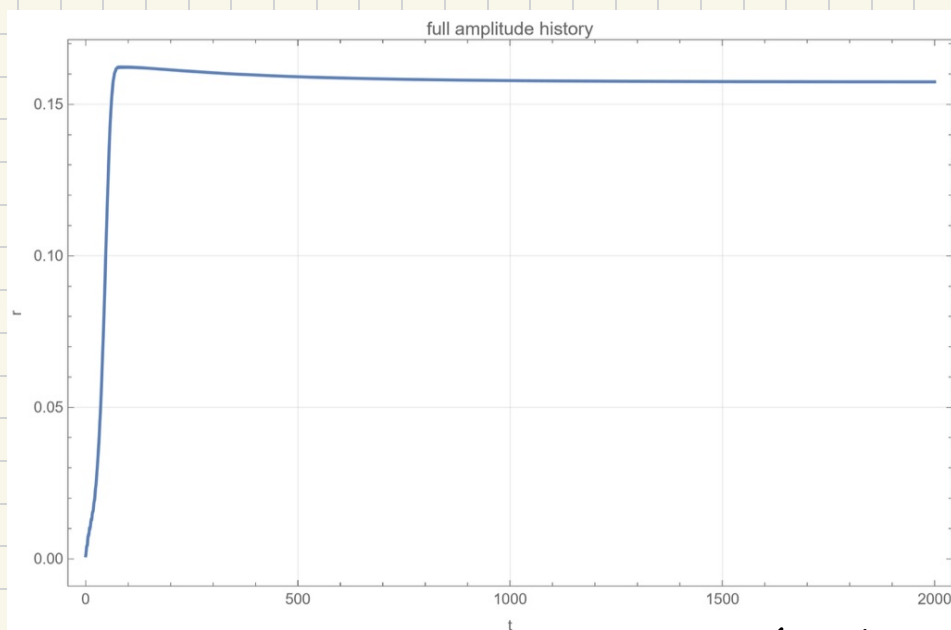
我選的參數下 $\rho = 1.0 > 1 \Rightarrow$ 零振幅平衡存在, 但未必代表為吸引態 ✖

(c) 有 (如(d)之邊臨界可知)

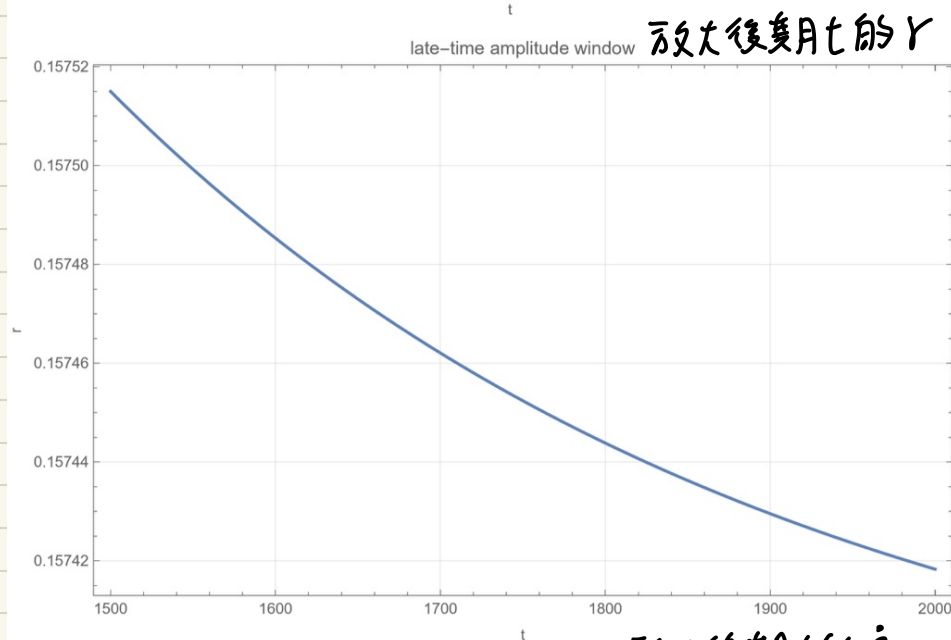
此系統雖有阻尼, 但不是封閉耗能系統 (因圓盤由外部馬達維持穩定轉速 ω), 馬達可持續注入能量支撐阻尼消耗能量, 以達成動態平衡, 使系統不收敛而進入非平衡長時間週期運動

$\begin{cases} \text{if } \gamma \rightarrow \text{const. 且 } \dot{\beta}_1 \rightarrow 0 \text{ 且 } \dot{\beta}_2 \rightarrow 0 \Rightarrow \text{有靜態平衡解} \\ \text{if } \gamma \text{ 持續振盪 且 } \dot{\beta}_i \neq 0 \Rightarrow \text{滾珠持續相對圓盤滾動} \\ \Rightarrow \text{進入非平衡長時間週期運動} \end{cases}$

for $\omega = 0.8 < \omega_n$



$Y \rightarrow \text{const.}$

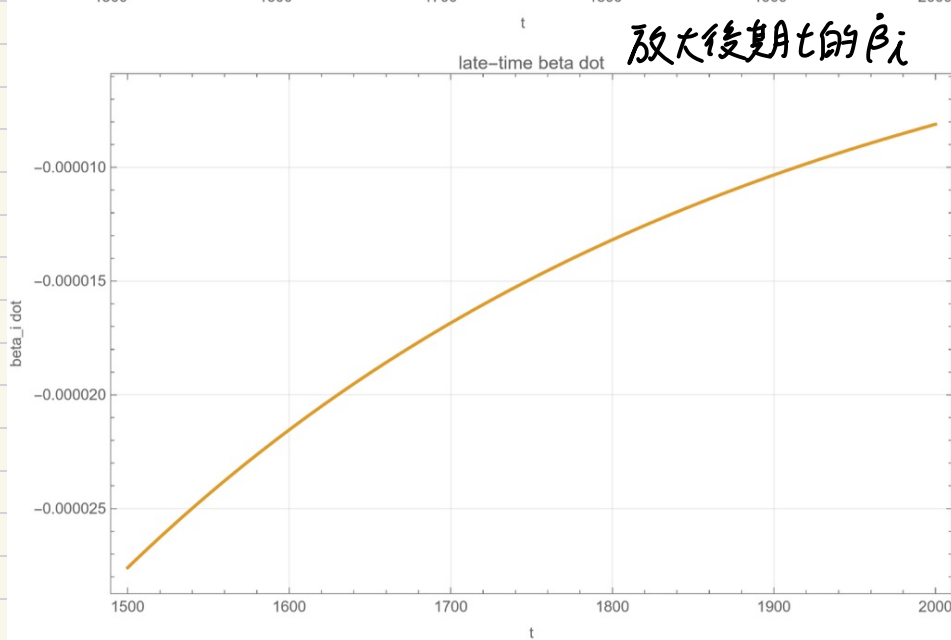


放大後變化的 Y

Y 下降數值很小

$\Downarrow$

$Y \rightarrow \text{const.}$

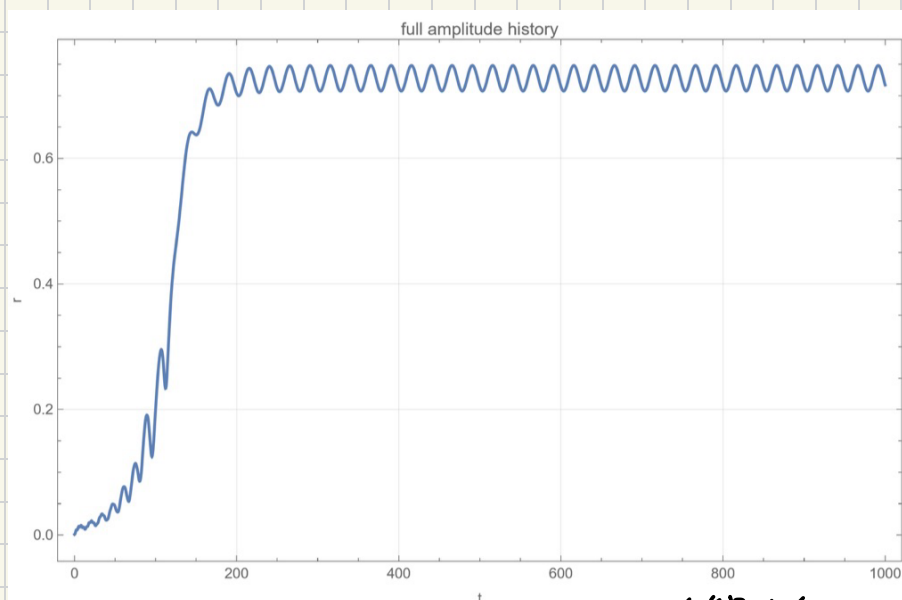


放大後變化的 β_i

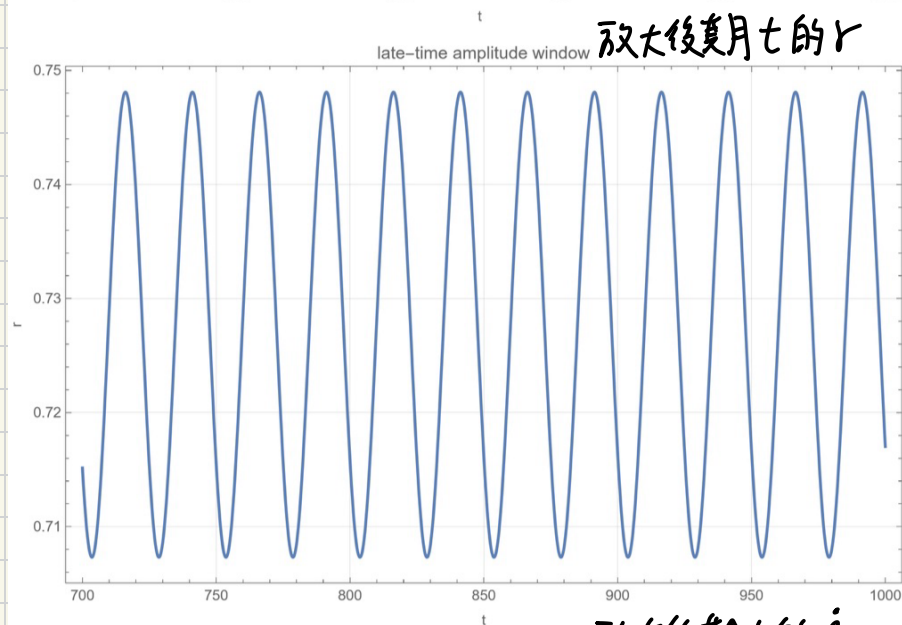
$\beta_i \dot \rightarrow 0$ (數值很小)

$\Rightarrow$ 是靜態平衡了

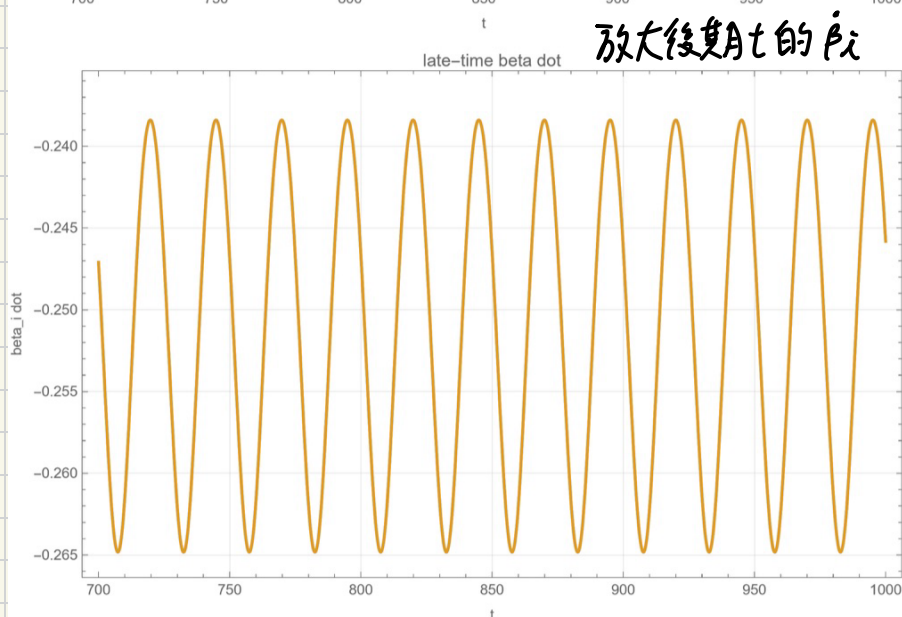
for $\omega = 1.2 > \omega_n$



$r \rightarrow \text{const.}$
在 0.73 附近振盪



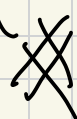
$r \rightarrow \text{const.}$
在 0.73 附近振盪



$\beta_i \rightarrow 0$

$\Rightarrow$ 滾珠持續相對圓盤滾動

$\Rightarrow$ 存在除了平穩解以外, 為週期性長時間運動的運動形式



Q2Cd)(e) 用 Mathematica code.

```
ClearAll["Global`*"];
```

清除全部

```
WrapAngle[theta_] := Mod[theta + Pi, 2 Pi] - Pi;
```

模除

π

π

π

```
SetOptions[{Plot, ParametricPlot, ListLinePlot},
```

指派選項

繪圖

參數繪圖

點集的線繪圖

```
Frame → True,
```

邊框

真

```
GridLines → Automatic,
```

格線

自動

```
PlotRange → All,
```

繪製範圍

全部

```
ImageSize → 620
```

影像尺寸

```
];
```

```
q2DefaultParams = <|
```

```
"M" → 1.0,
```

```
"me" → 0.05,
```

```
"mb" → 0.05,
```

```
"e" → 0.2,
```

```
"b" → 1.0,
```

```
"k" → 1.15,
```

```
"c" → 0.05,
```

```
"cb" → 0.05
```

```
|>;
```

```
Q2RHS[z_List, omegaVal_?NumericQ, p_ : q2DefaultParams] := Module[
```

清單

數值量判定

模組

```
{xv, yv, beta1v, beta2v, xdv, ydv, beta1dv, beta2dv,
```

```
Mv, mev, mbv, ev, bv, kv, cv, cbv, mv, Gv,
```

```
S1v, C1v, S2v, C2v, Om1v, Om2v, mat, rhs, qdd},
```

```
{xv, yv, beta1v, beta2v, xdv, ydv, beta1dv, beta2dv} = z;
```

```
Mv = p["M"]; mev = p["me"]; mbv = p["mb"];
```

```
ev = p["e"]; bv = p["b"]; kv = p["k"]; cv = p["c"]; cbv = p["cb"];
```

```
mv = Mv + mev + 2 mbv;
```

```
Gv = mbv bv;
```

```
S1v = Sin[beta1v]; C1v = Cos[beta1v];
```

正弦

餘弦

```
S2v = Sin[beta2v]; C2v = Cos[beta2v];
```

正弦

餘弦

```
Om1v = omegaVal + beta1dv;
```

```
Om2v = omegaVal + beta2dv;
```

```
mat = {
```

```
{mv, 0, -Gv S1v, -Gv S2v},
```

```
{0, mv, Gv C1v, Gv C2v},
```

```

{-Gv S1v, Gv C1v, mbv bv^2, 0},
{-Gv S2v, Gv C2v, 0, mbv bv^2}
};

rhs = {
  2 mv omegaVal ydv + mv omegaVal^2 xv + mev ev omegaVal^2
  + Gv (Om1v^2 C1v + Om2v^2 C2v) - kv xv - cv (xdv - omegaVal yv),
  -2 mv omegaVal xdv + mv omegaVal^2 yv
  + Gv (Om1v^2 S1v + Om2v^2 S2v) - kv yv - cv (ydv + omegaVal xv),
  -Gv ((2 omegaVal xdv - omegaVal^2 yv) C1v + (2 omegaVal ydv + omegaVal^2 xv) S1v)
  - cbv bv^2 beta1dv,
  -Gv
  ((2 omegaVal xdv - omegaVal^2 yv) C2v + (2 omegaVal ydv + omegaVal^2 xv) S2v)
  - cbv bv^2 beta2dv
};

qdd = LinearSolve[mat, rhs];
      |線性求解
Join[{xdv, ydv, beta1dv, beta2dv}, qdd]
      |連接
];

Q2DefaultInitialCondition[p_ : q2DefaultParams] := Module[{Fv, Gv, alpha},
      |模組
  Fv = p["me"] p["e"];
  Gv = p["mb"] p["b"];
  alpha = ArcCos[Fv / (2 Gv)];
      |反餘弦
  {10^-3, 0., Pi + alpha + 0.05, Pi - alpha - 0.05, 0., 0., 0., 0.}
      |π |π
];

Q2Solve[omegaVal_?NumericQ, tmax_ : 400,
      |數值量判定
  p_ : q2DefaultParams, z0_ : Automatic] := Module[
      |自動 |模組
  {vars, init},
  vars = {xx, yy, beta1, beta2, vx, vy, vbeta1, vbeta2};
  init = If[z0 == Automatic, Q2DefaultInitialCondition[p], z0];
      |如果 |自動
  NDSolveValue[
      |常微分方程解數值化
  Join[
      |連接
    Thread[D[Through[vars[t]], t] == Q2RHS[Through[vars[t]], omegaVal, p]],
      |執行緒 |... |貫穿 |貫穿
    Thread[Through[vars[0]] == init]
      |執行緒 |貫穿
  ],
  vars,
  {t, 0, tmax},
  MaxSteps -> Infinity,
      |最大步數 |無窮大

```

```

MaxStepFraction → 1 / 10000,
步長最大比例
AccuracyGoal → 10,
精確度目標
PrecisionGoal → 10
精密目標
]
];

Q2RequiredFigure[sol_List, omegaVal_?NumericQ, tmax_?NumericQ] := Module[
清單 數值量判定 數值量判定 模組
{
xv, yv, beta1v, beta2v, beta1dv, beta2dv, rfun},
xv = sol[[1]]; yv = sol[[2]]; beta1v = sol[[3]]; beta2v = sol[[4]];
beta1dv = sol[[7]]; beta2dv = sol[[8]];
rfun[tt_] := Sqrt[xv[tt]^2 + yv[tt]^2];
平方根
Column[{
欄
Style[
樣式
"Q2(d) corrected required plots, omega = " <> ToString[omegaVal], 15, Bold],
轉換成字串 粗體
Plot[rfun[t], {t, 0, tmax},
繪圖
PlotLabel → "amplitude r(t)", FrameLabel → {"t", "r"}],
繪製標籤 邊框標籤
Plot[Evaluate[{WrapAngle[beta1v[t]], WrapAngle[beta2v[t]]}], {t, 0, tmax},
繪圖 計算
PlotLabel → "wrapped ball angles beta_i(t)",
繪製標籤
FrameLabel → {"t", "wrapped beta_i"},
邊框標籤
PlotLegends → {"beta1", "beta2"}],
繪製圖例
Plot[Evaluate[{beta1dv[t], beta2dv[t]}], {t, 0, tmax},
繪圖 計算
PlotLabel → "ball relative angular velocities beta_i dot",
繪製標籤
FrameLabel → {"t", "beta_i dot"},
邊框標籤
PlotLegends → {"beta1dot", "beta2dot"}]
繪製圖例
}, Spacings → 1.0]
間隔
];

Q2TrajectoryFigure[sol_List, omegaVal_?NumericQ, tmax_?NumericQ] :=
清單 數值量判定 數值量判定
Module[{xv, yv},
模組
xv = sol[[1]]; yv = sol[[2]];
Column[{
欄

```

```

Style["Q2 supplemental trajectory diagnostic, omega = " <>
  ToString[omegaVal], 15, Bold],
  ToString[omegaVal], 15, Bold],
  ParametricPlot[Evaluate[{xv[t], yv[t]}], {t, 0, tmax},
    PlotLabel → "trajectory of disk center C in rotating coordinates",
    FrameLabel → {"x", "y"}],
  Plot[Sqrt[xv[t]^2 + yv[t]^2], {t, 0, tmax},
    PlotLabel → "amplitude r(t)", FrameLabel → {"t", "r"}],
  Spacings → 1.0]
];

Q2LongTimeFigure[sol_List,
  omegaVal_?NumericQ, t0_?NumericQ, t1_?NumericQ] := Module[
  {xv, yv, beta1dv, beta2dv, rfun},
  xv = sol[[1]]; yv = sol[[2]]; beta1dv = sol[[7]]; beta2dv = sol[[8]];
  rfun[t_] := Sqrt[xv[t]^2 + yv[t]^2];
  Column[{
    Style["Q2(e) long-time diagnostic, omega = " <> ToString[omegaVal], 15, Bold],
    Plot[rfun[t], {t, 0, t1},
      PlotLabel → "full amplitude history", FrameLabel → {"t", "r"}],
    Plot[rfun[t], {t, t0, t1},
      PlotLabel → "late-time amplitude window", FrameLabel → {"t", "r"}],
    Plot[Evaluate[{beta1dv[t], beta2dv[t]}], {t, t0, t1},
      PlotLabel → "late-time beta dot", FrameLabel → {"t", "beta_i dot"},
      PlotLegends → {"beta1dot", "beta2dot"}],
    Spacings → 1.0]
];

q2SolSub = Q2Solve[0.8, 2000];

```

```
q2ReqSub = Q2RequiredFigure[q2SolSub, 0.8, 400];  
q2ReqSub
```

```
q2TrajSub = Q2TrajectoryFigure[q2SolSub, 0.8, 2000];  
q2TrajSub
```

```
q2LongSub = Q2LongTimeFigure[q2SolSub, 0.8, 1500, 2000];  
q2LongSub
```

```
q2SolSuper = Q2Solve[1.2, 1000];
```

```
q2ReqSuper = Q2RequiredFigure[q2SolSuper, 1.2, 400];  
q2ReqSuper
```

```
q2TrajSuper = Q2TrajectoryFigure[q2SolSuper, 1.2, 1000];  
q2TrajSuper
```

```
q2LongSuper = Q2LongTimeFigure[q2SolSuper, 1.2, 700, 1000];  
q2LongSuper
```